Reputational Bargaining with Espionage

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Abstract

We study how spying opportunities influence bilateral bargaining outcomes when negotiating parties possess private information about their reservation values. In a continuous-time war of attrition, parties have the opportunity to invest in espionage technology to receive information about whether the opponent is obstinate regarding their claimed reservation value. The potential arrival of information incentivizes the players to wait, thereby influencing both the reputation-building process and the settlement. The evolution of reputations and strategies is characterized by Riccati differential equations and is solved analytically. When the space of demand is sufficiently rich and the prior probability of being obstinate is vanishingly small, the negotiation concludes almost immediately, and the party with the capability to learn captures the entire negotiation surplus, reminiscent of the Coase conjecture.

Keywords: Bargaining, reputation, information acquisition, war of attrition

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1 Introduction

In many negotiations, the involved parties rely on espionage activities to gain information about their opponents and thereby secure more favorable outcomes. In commercial negotiations, firms often attempt to obtain intelligence about an opponent's proposal strategy, reservation value, or even whether the opponent behaves obstinately.[2] Such information can affect beliefs about the counterpart's willingness to concede and thereby shape the agenda and the final outcome of the negotiation. Similar information-gathering practices also arise in diplomatic and international trade negotiations, where states often rely on intelligence resources to assess their opponent's bargaining power in advance and to calibrate their negotiation strategies accordingly.

Despite its prevalence in real-world negotiations, the role of information acquisition in dynamic bargaining remains underexplored, particularly when reputational concerns are present. As a strategic instrument, a central question about espionage is whether such information acquisition is indeed effective. The potential benefit of espionage lies in its ability to guide strategic adjustment: when an informed rational party learns that the opponent is obstinate, it may concede immediately and avoid unnecessary delay costs; when the informed party discovers that the opponent is strategic, it may instead choose to delay strategically in anticipation of the opponent's eventual concession. Moreover, when espionage activities are unverifiable, the mere capability to acquire information can accelerate reputation formation, making the informed party appear more resolute and enhancing its bargaining power.

This paper studies how information acquisition shapes the evolution and outcome of reputational bargaining. Specifically, we address two questions. First, how does the opportunity for information acquisition affect strategic behavior, the bargaining power of the involved parties, and the distribution of the total surplus? Second, how valuable is information when both parties' commitment power becomes vanishingly small?

Our model extends the infinite-horizon reputational bargaining framework of Abreu and Gul (2000) (henceforth AG) by introducing information acquisition. Two players, denoted by 1 ("he") and 2 ("she"), bargain over the division of a unit-sized pie. Each player is either a behavioral type or a rational type. A behavioral player demands a fixed share of the pie and never accepts any offer below this demand, whereas a rational player can make any demand and accept any offer. At the outset, players sequentially announce their demands. Subsequently, negotiations proceed as each player maintains the announced demand, and information arrives randomly over time.

[2]For example, during the 2009 iron-ore price negotiations, several Rio Tinto employees were detained and later convicted in China for obtaining confidential information from Chinese steelmakers (The Guardian, 2009). L'Oréal was likewise found to have relied on technical information obtained during its negotiation talks with Olaplex (Los Angeles Times, 2019). In the 2024 Phillips 66 case, a California court concluded that the company had obtained Propel Fuels' commercial data during due-diligence talks, resulting in a $800 million damages award (Reuters, 2024).

Upon the arrival of information, the informed player privately learns the opponent's type.

Learning the opponent's type helps a party assess whether the opponent will stand firm on their demands. In union-management bargaining, employers often infer the union's obstinacy by observing the financial reserves and the level of public support. In M&A negotiations, acquirers rely on consultants or industry analysis to learn about the target firm's liquidity constraints and urgency to sell, i.e., information that helps determine whether the opponent is rigid or strategically flexible. In war and diplomatic negotiations, intelligence and reconnaissance activities serve to assess whether the opposing side is resolute or can be deterred and persuaded to compromise.

In the AG model, a fraction of players are behavioral types who never concede from their stated demands. Rational players have incentives to mimic these types to build a reputation for toughness and induce their opponents to accept their offers. Because neither player can acquire information about the other's type, reputation evolves solely through the pattern of concessions. In the unique dynamic equilibrium with a single behavioral–type space, players announce their demands at the outset, and at most one player concedes with positive probability at time zero. Thereafter, both players concede at constant hazard rates, causing reputations to rise exponentially over time. This process continues until both reputations reach one, by which point all rational players have exited and only behavioral types remain.

We begin with the case in which one party can employ espionage technology at zero cost.[3] If initial negotiation efforts fail, information about Player 2's type is revealed to Player 1 according to a Poisson rate γ_1. Upon receiving the information, Player 1 privately learns whether Player 2 is of the behavioral type. The ability γ_1 is common knowledge and exogenously given.[4] Whether Player 1 actually learns Player 2's type, however, is privately observed.

We first study the case in which each player may have only one possible behavioral demand. In Theorem 1, we characterize the unique equilibrium. A rational Player 1 who learns that Player 2 is behavioral immediately concedes to avoid unnecessary delay costs. Because the spying activity is private, Player 1 cannot credibly disclose the obtained evidence to terminate the negotiation directly. Conversely, when Player 1 learns that Player 2 is also rational, he strategically delays and holds firm to his demand. This delay accelerates the process of reputation building and strengthens his bargaining position, eventually inducing the opponent to concede. Proposition 2 establishes that an improvement in Player 1's information acquisition intensity (γ_1) strengthens

[3]This simplification is without loss of generality in environments where the marginal cost of intelligence gathering is negligible relative to the stakes of the interaction. For example, in military conflicts, the cost of espionage is typically negligible compared to the overall cost of warfare itself; in large-scale merger negotiations, the cost of obtaining proprietary information, such as due diligence teams or external advisors, is negligible relative to the economic value at stake in the transaction. Theorem 3 discusses the extension where both players can acquire information.

[4]In Section 5, we discuss the case in which the received information is not entirely accurate; the analytical conclusions remain consistent.

his bargaining position, leading to a higher equilibrium payoff for Player 1 and a lower one for Player 2. In line with AG, Proposition 3 further demonstrates that players' patience plays a role in the distribution of the surplus.

We then focus on the limit case where both players' commitment power vanishes. When the prior probability of behavioral types becomes vanishingly small, Player 2 concedes immediately at time zero, allowing Player 1 to obtain his demanded share instantly (Proposition 4). This result echoes the Coase conjecture, which asserts that in dynamic bargaining with continuous offers, if the buyer possesses full informational advantage while the seller lacks commitment power, the seller's surplus dissipates and the buyer captures the entire surplus. We show that even in a two-sided incomplete information setting, information acquisition plays an important role in determining the distribution of the total surplus.

Through our equilibrium characterization, we address the first motivating question of how information acquisition shapes strategic behavior. For the informed Player 1, acquiring information eliminates uncertainty about the opponent's type, either avoiding unnecessary waiting costs or increasing the continuation value of waiting. Upon learning that Player 2 is rational, Player 1 strictly prefers to wait rather than concede, thereby building a tougher reputation that eventually induces Player 2's concession. Because Player 1 may become informed over time and, once informed, strictly prefers to wait, the concession rate that keeps an uninformed Player 1 indifferent must fall. Thus, Player 2 concedes with a lower probability at each moment compared to AG. By contrast, Player 2 cannot acquire information about Player 1 and therefore continues to concede at the same constant hazard rate as in AG. Consequently, Player 1's overall concession rate remains unchanged, while Player 2's declines, shortening Player 1's reputation-building phase and lengthening Player 2's.

The second question concerns how information acquisition affects the division of surplus. When both players' commitment powers vanish, information acquisition becomes decisive: it induces the uninformed Player 2 to concede immediately to Player 1's demand, regardless of their relative initial reputation. An informed Player 1 who learns that Player 2 is rational strategically delays to strengthen his reputation. Information acquisition, therefore, accelerates reputation formation. Relative to AG, the time required for Player 1's reputation to reach one is shortened, while Player 2's reduced concession rate lengthens his own reputation-building process and increases the likelihood of conceding at time zero. When the prior probability of behavioral types vanishes for both players, reputations in AG take infinitely long to form, and relative commitment powers determine payoffs. In contrast, with information acquisition, Player 1's reputation always forms in finite time whereas Player 2's remains infinite. Consequently, Player 2 must concede almost fully at time zero to sustain the equilibrium timing, and Player 1's demand is immediately satisfied.

In Theorem 2, we extend the analysis to the case in which both players possess rich behavioral type spaces. The equilibrium outcome remains the same. When both players' initial reputations converge to zero, information acquisition also renders the negotiation efficient. Since Player 2 concedes almost immediately at time zero, Player 1 optimally chooses the most aggressive demand. As a result, Player 1 captures nearly the entire negotiation surplus, regardless of the intensity of information acquisition.

We also examine the endogenous choice of learning intensity, allowing Player 1 to strategically invest in information acquisition. The cost of information acquisition increases convexly with the intensity of information arrival. Because spying accelerates reputation formation, a rational Player 1 invests in stronger information technology when Player 2's initial reputation is sufficiently low, thereby increasing the likelihood of inducing an immediate concession. When Player 2 already appears highly resolute, additional intelligence yields little marginal benefit, and Player 1 optimally refrains from acquiring information. This endogenous choice produces a sharp comparative static: spying is valuable when the opponent's commitment is uncertain but becomes ineffective once the opponent already appears obstinate.

We further analyze a bilateral information acquisition environment in which players sequentially and endogenously choose their information-acquisition intensities prior to bargaining in Theorem 3. The equilibrium structure reveals that information acquisition is valuable only for players who expect not to concede at the outset of bargaining. Players who anticipate conceding immediately optimally avoid information acquisition once it is costly. As a result, equilibrium behavior features information acquisition by at most one player. This insight provides a rationalization for the one-sided information acquisition structure adopted in the baseline model, showing that it arises endogenously from reputational incentives rather than from an exogenous modeling restriction.

Our model applies to negotiation environments in which one party can secretly acquire information about the opponent's resolve. Such situations commonly arise in mergers and acquisitions, labor disputes, and diplomatic negotiations, where firms, unions, or states use intelligence or private channels to assess whether the counterpart will stand firm or eventually concede. We show that information acquisition matters not only because it reduces uncertainty about the opponent's type, but also because it strengthens the informed party's reputation. This mechanism provides a complementary explanation for why, in practice, private intelligence often translates into greater bargaining power and faster settlements.

Related literature

Dynamic reputational bargaining studies how delays and surplus division arise when players may be committed to fixed demands. Early contributions establish the core war-of-attrition structure. Chatterjee and Samuelson (1987) analyze a discrete-time concession game with two-sided incomplete information, and Myerson (1991) formalizes one-sided reputational dynamics. The canonical formulation of Abreu and Gul (2000) characterizes the unique continuous-time equilibrium with two-sided incomplete information and shows its convergence from discrete-time bargaining. Their framework anchors subsequent work in which reputations evolve through concessions.

Subsequent work refines this structure by examining how reputational uncertainty and small commitment probabilities disciplines equilibrium behavior. Wolitzky (2012) shows that the maxmin share a player can guaranty under first-order rationality can be large relative to the commitment probability. Abreu et al. (2015) study one-sided uncertainty about patience via behavioral perturbations and show that dynamic perturbations generate mixed strategies and persistent delays as patient types separate by waiting. Fanning (2018) shows that uncertainty about future delay costs can sustain substantial deadlock even as behavioral types vanish since rational agents strategically demand nearly the entire surplus and wait for news. Sanktjohanser (2023) analyzes that once behavioral types can choose their initial demands, any feasible rational payoff can be sustained when commitment probabilities are small. Across these studies, reputations evolve primarily through concession patterns, and players do not actively acquire information.

A complementary strand of the literature explores how reputation interacts with additional sources of institutional design. Lee and Liu (2013) consider repeated interactions in which a long-lived player strategically chooses gamble actions to build a reputation for commitment through stochastic payoff realizations. Fanning (2023) analyzes mediated reputational bargaining and shows that mediators improve outcomes only under specific conditions, whereas arbitrators may perform better but can impose disagreements. Relatedly, Fanning (2021) shows that noisy mediator communication can raise the payoffs of rational bargainers; Ekmekci and Zhang (2025) demonstrate that external resolution or ultimatum opportunities can either benefit or harm an unjustified party, with immediate efficient agreement emerging as justification probabilities vanish; and Pei and Vairo (2025) link reputation formation to technology adoption, showing that reputation incentives can generate inefficient under-adoption when bargaining delays are prolonged. In our setting, the possibility of privately learning introduces an additional channel through which information arrival influences reputation formation and concession dynamics.

Information arrival and endogenous learning have also been studied directly in dynamic environments. Kim and Lee (2014) examine a war of attrition in which agents may learn a payoff-relevant state before a stochastic deadline. Unverifiable learning induces strategic delay and re-

dundant information acquisition; however, with sufficiently small costs, these distortions cancel out in symmetric equilibrium. Vasserman and Yildiz (2019) analyze pretrial negotiation with anticipated public information, showing that news arrival triggers immediate settlement and that, before arrival, settlement timing depends on optimism, ranging from immediate agreement to deadline settlement or impasse.

Existing work also studies how information acquisition shapes bargaining power in static bilateral trade. Solan and Yariv (2004) study normal-form games in which a player can purchase noisy information about an opponent's future responses and show the information purchased is independent of its cost when the game is non-degenerate. Roesler and Szentes (2017) characterize buyer-optimal signal structures under take-it-or-leave-it offers and show that all such structures implement efficient trade and a common unit-elastic demand. Ravid et al. (2022) extend this to endogenous information acquisition and show that as learning costs vanish, equilibrium outcomes converge to the worst free-learning equilibrium. Chatterjee et al. (2025) show that in an ultimatum setting, the mere option to acquire a confirmatory signal allows the buyer to extract the entire surplus, as information becomes arbitrarily cheap, even though no information is ever acquired on the equilibrium path.

Beyond bargaining, mechanism-design environments with endogenous information acquisition also highlight implications. Shi (2012) shows that auctions with pre-play information gathering require lower optimal monopoly prices and modified reserve prices relative to Myerson's benchmark. In winner-take-all contests, Chen (2025) shows that acquiring private information about rivals' strengths improves performance without increasing effort, though learning may not occur when signal accuracies are unobservable.

Our analysis also relates to the Coase conjecture and its extensions (Fudenberg et al., 1987; Gul et al., 1986; Myerson, 1991). These classic contributions study dynamic bargaining with one-sided asymmetric information—typically with an uninformed seller and a privately informed buyer—and show that the seller obtains essentially no surplus in the limit. Abreu and Gul (2000) extend this insight to two-sided incomplete information through a reputational war-of-attrition framework. In our two-sided environment, Coasian forces re-emerge in an asymmetric form: as commitment probabilities vanish, the party capable of privately acquiring information about the opponent captures the entire surplus.

The remainder of the paper proceeds as follows. Section 2 presents the baseline model. Section 3 characterizes equilibrium behavior and shows how information alters concession dynamics and the evolution of beliefs. Section 4 examines the limiting environment in which commitment probabilities vanish. Section 5 discusses extensions of our model and concludes.

2 Model

We focus here on a simplified version of our general model. Consider a dynamic bargaining game between two players $i \in \{1, 2\}$ who negotiate over a unit pie. We begin with a baseline specification in which each player has a single possible behavioral type. A player is either rational or behavioral. A behavioral player i is committed to demanding a fixed share $a_i \in (0, 1)$ of the pie and never accepts any offer below this demand. A rational player can choose any demand and may concede at any time. The pair of demands (a_1, a_2) satisfies $a_1 + a_2 > 1$, implying a positive disagreement surplus $d = a_1 + a_2 - 1 > 0$. The prior probability that Player i is behavioral is $z_i \in (0, 1)$, and these priors are common knowledge.

Time is continuous and extends to infinity. Player i discounts future payoffs at rate $r_i > 0$. At the outset, Player 1 is endowed with an information–acquisition capacity summarized by an intensity parameter $\gamma_1 \geq 0$. In the baseline model, this information intensity is exogenously given and only Player 1 has access to information acquisition. Types are then realized according to the common prior. Player 1 announces a demand a_1, which Player 2 observes. Player 2 either accepts the implied offer $1 - a_1$ or makes a counter-demand $a_2 > 1 - a_1$. If agreement is not reached immediately, the interaction proceeds to a continuous-time war-of-attrition subgame in which, at every instant $t > 0$, each player may concede. The first concession terminates the game and implements the opponent's most recent proposal; if both players concede simultaneously, each proposal is implemented with probability $1/2$.

During the continuous-time bargaining phase, information arrives according to a Poisson process with rate γ_1. Upon arrival, Player 1 privately observes a signal that perfectly reveals Player 2's type. Information acquisition is costless and one-sided in the baseline specification: Player 2 receives no signals. Thus, learning occurs only through Player 1's stochastic information arrival while strategic concessions unfold over time.

This timing captures environments in which information capacity must be determined prior to the realization of specific bargaining needs. In many institutional settings, organizations commit ex ante to intelligence or consulting expenditures, while the strategic value of that information materializes only once a concrete negotiation emerges.

To summarize, the baseline bargaining game $B = (a_1, a_2, z_1, z_2, r_1, r_2, \gamma_1)$ is characterized by single behavioral demands a_1 and a_2; prior probabilities z_1 and z_2; discount rates r_1 and r_2; and an exogenous one-sided spying intensity γ_1 available only to Player 1. Section 4 generalizes the framework to accommodate richer behavioral type spaces and endogenous, bilateral information acquisition.

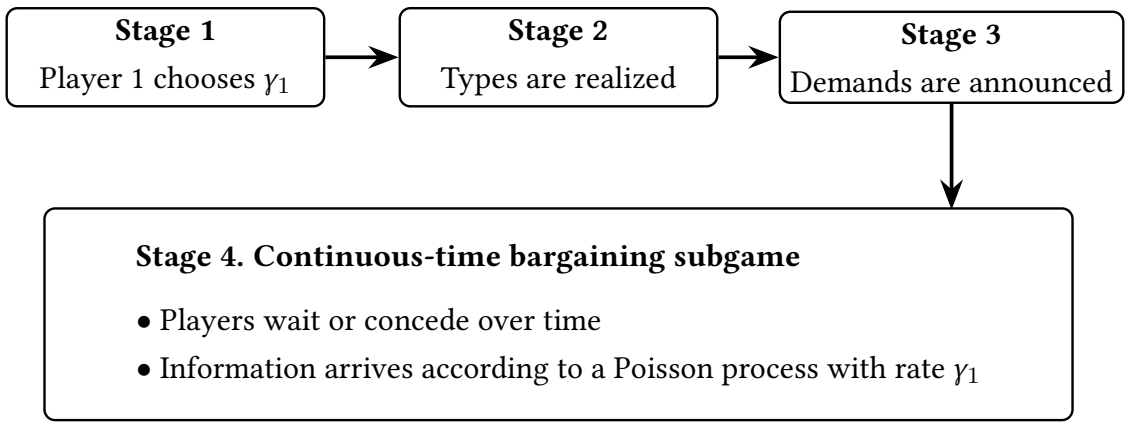


Figure 1: Timing of the bargaining game with one-sided information acquisition

Strategies and utility

Since we start with the case of a single behavioral demand for each player and zero spying costs, the strategy space is reduced to deciding at what time to concede once players have announced their respective demands. Let $F_i(t)$ denote the cumulative probability that a rational player i has conceded by time t, and $\mu_i(t)$ denote the opponent's belief at time t that Player i will never concede. After the announcement stage, rational players' strategy profiles (σ_1, σ_2) can thus be represented by (F_1, F_2).

A rational player i's expected utility from conceding at time t is

$$\begin{aligned} v_i(t, \sigma_j) = (1 - z_j) \int_0^t a_i e^{-r_i s} dF_j(s) + \left[1 - (1 - z_j) F_j(t)\right] e^{-r_i t} (1 - a_j) \\ + (1 - z_j) \left[F_j(t) - F_j(t^-)\right] \frac{a_i + 1 - a_j}{2}, \end{aligned} \tag{1}$$

where $F_j(t^-) := \lim_{s \uparrow t} F(s)$. Player i's expected utility from strategy σ_i is

$$u_i(\sigma_i, \sigma_j) = \int_0^\infty v_i(s, \sigma_j) dF_i(s).$$

We will describe the evolution of both players' reputations in Section 3.

Our equilibrium concept is a *Perfect Bayesian Equilibrium* (PBE) defined by Fudenberg and Tirole (1991). A strategy profile (F_1, F_2) and a belief system (μ_1, μ_2) constitute a PBE if, given the belief system, each player's concession strategy F_i maximizes his or her expected discounted payoff, and beliefs are updated consistently with Bayes' rule along the equilibrium path.

3 Equilibrium

We first characterize the equilibrium with a single behavioral demand for each player, then we discuss equilibrium strategies and reputation evolution.

3.1 Equilibrium characterization

Theorem 1. *Consider a bargaining game with one-sided information acquisition opportunity and single demand types $B = (a_1, a_2, z_1, z_2, r_1, r_2, \gamma_1)$. There exists a unique equilibrium. There exists time T such that every equilibrium strategy profile (F_1, F_2) satisfies the following properties.*

1. *Rational Player 1 concedes immediately after he knows that Player 2 is behavioral, and strictly prefers waiting after he knows that Player 2 is rational, while uninformed rational Player 1 is indifferent between conceding and waiting.*

2. *F_1 and F_2 are atomless and strictly increasing in $(0, T^*]$, and at most one of them has an atom at $t = 0$;*

3. *Both players' reputations simultaneously reach one at time T, and no rational players stay in the bargaining for $t > T$.*

For the uninformed players, the proof follows arguments similar to those in AG. The key observation is that an uninformed player remains indifferent between waiting and conceding throughout the bargaining process. Property 1 states that once information arrives, the informed Player 1 concedes immediately upon learning that Player 2 is behavioral, thereby avoiding unnecessary delay costs since Player 2 never concedes. However, upon learning that Player 2 is rational, Player 1 then strictly prefers to wait. The intuition is straightforward: once uncertainty about the opponent's type is resolved, Player 1's continuation value in equilibrium becomes strictly greater than the immediate payoff $1 - a_2$ from conceding. Consequently, Player 1 chooses to wait until time T given no concession from Player 2, at which point rational Player 2 eventually concedes.

Properties 2 and 3 share similar features with those in AG. Property 2 states that within any subinterval of $(0, T]$, both rational players concede with positive probability. As established in Property 1, the informed Player 1 strictly prefers to delay, while the uninformed players remain indifferent between conceding and waiting, which ensures that the cumulative concession probabilities are strictly increasing over time. Who concedes with positive probability at time zero depends on the relative speed of reputation building: the player whose reputation reaches one more slowly must concede with positive probability at time zero so that both reputations reach one simultaneously. Property 3 further confirms that all rational players exit the bargaining

within a finite time T. Rational Player 2 must concede before T, leaving only behavioral types in the game thereafter.

Building on these three properties, we can derive the closed-form solutions for the rational players' equilibrium strategies. In the next section, we first characterize both players' concession rates. Using these concession rates together with the information arrival rate γ_1, we then derive the evolution of players' reputations, which allows us to compute their equilibrium payoffs.

3.2 Equilibrium strategies

Property 1 shows that the informed Player 1 strictly prefers to delay. This implies that the continuation value from waiting must be strictly greater than the immediate payoff from conceding:

$$\mathbb{E}[u_1 \mid \mu_1(t)] = \lambda_2(t)dt \cdot a_1 + [1 - \lambda_2(t)dt]\, e^{-r_1 dt} \cdot \mathbb{E}[u_1 \mid \mu_1(t + dt)] > 1 - a_2,$$

in $[0, T)$, where $\lambda_2(t)$ denotes the concession rate of a rational Player 2 at time t. We use $U_1(t)$ as a shorthand for $\mathbb{E}[u_1 \mid \mu_1(t + dt)]$. As $dt \to 0$, the above equation can be rearranged to

$$U_1'(t) = [\lambda_2(t) + r_1]U_1(t) - \lambda_2(t)a_1. \tag{2}$$

Property 2 also implies that, the uninformed Player 1 is indifferent between conceding and waiting in $(0, T]$. Hence, the continuation value of waiting must equal the payoff from immediate conceding. Let $\mu_2(t)$ denote Player 1's belief at time t that Player 2 will never concede. Then Player 2's concession rate $\lambda_2(t)$ must adjust to maintain this indifference:

$$\begin{aligned}&(1 - \mu_2(t))\lambda_2(t)dt \cdot a_1 + [1 - (1 - \mu_2(t))\lambda_2(t)dt]\, e^{-r_1 dt}[\gamma_1 dt \cdot \\ &(1 - \mu_2(t))\mathbb{E}[u_1 \mid \mu_1(t + dt)] + (1 - \gamma_1 dt(1 - \mu_2(t))) \cdot (1 - a_2)] = 1 - a_2.\end{aligned}$$

To interpret this equation, the left-hand side represents the expected payoff of the uninformed Player 1 from choosing to wait. If Player 1 decides to wait, his expected payoff arises from three sources: the probability that Player 2 concedes within the next infinitesimal interval dt; the continuation value if Player 1 privately receives information revealing that Player 2 is rational; and the expected value when Player 1 learns that Player 2 is behavioral and immediately concedes, or alternatively, when he decides to concede at time $t + dt$. The right-hand side represents the immediate payoff from conceding. As the time increment $dt \to 0$, the equality determines Player 2's equilibrium concession rate

$$\lambda_2(t) = \frac{r_1(1 - a_2) - \gamma_1(1 - \mu_2(t))(U_1(t) - (1 - a_2))}{(1 - \mu_2(t))(a_1 + a_2 - 1)} > 0. \tag{3}$$

Given that Player 1 strictly prefers to wait after learning that Player 2 is rational, Player 2's concession rate differs slightly from that in AG. In our model, the concession rate of Player 2 is smaller than that in AG.

Because Player 2 can build a reputation only through the concessions of her rational type, her reputation increases more slowly than in AG. By Bayes' rule, conditional on the game not having ended by time t,

$$\mu_2(t) = \frac{z_2}{z_2 + (1 - z_2)\left[1 - F_2(t^-)\right]}.$$

The evolution of Player 1's belief about Player 2's type satisfies

$$\frac{\mu_2'(t)}{(1 - \mu_2(t))(\mu_2(t))} = \lambda_2(t). \tag{4}$$

To solve this system of equations, we require two boundary conditions. First, assume that Player 2 does not concede at time zero; her initial reputation equals her prior probability of being a behavioral type, that is, $\mu_2(0) = z_2$. Let T_2 denote the time at which Player 2's reputation would reach one in the absence of an initial concession. All rational types of Player 2 have exited the game at T_2, and an informed player 1's continuation value satisfies $U_1(T_2^+) = 1 - a_2$.

Solving this system of equations is algebraically cumbersome, and the detailed derivations are therefore provided in the Appendix. It is worth noting that this procedure uniquely determines the time T_2 at which Player 2's reputation reaches one, given that she does not concede with a positive probability at time zero. Due to the exponential structure, a closed-form expression for T_2 is generally not available; nevertheless, the implicit characterization is sufficient for our subsequent analysis.

Properties 1 and 2 imply that Player 2 is indifferent between conceding and waiting. Hence, as in AG, Player 1's overall concession rate must remain constant to maintain this indifference.

$$1 - a_1 = a_2 \cdot \widehat{\lambda}_1 dt + e^{-r_2 dt}(1 - a_1)(1 - \widehat{\lambda}_1 dt) \implies \widehat{\lambda}_1 = \frac{r_2(1 - a_1)}{a_1 + a_2 - 1}.$$

Since an informed Player 1 who learns that his opponent is rational strictly prefers to wait, the concession rate of the uninformed rational Player 1 is given by

$$\lambda_1(t) = \frac{r_2(1 - a_1)}{(1 - \mu_1(t))(a_1 + a_2 - 1)}.$$

3.3 Reputation evolution

We now characterize the evolution of players' reputations. Recall that, in our framework, reputation is defined as the opponent's belief that a player will never concede, rather than as the

belief that the player is of a behavioral type, as in much of the prior reputational bargaining literature. For Player 2, who cannot acquire information, reputation can increase only through the concessions of her rational type. The evolution of Player 2's reputation has been characterized above.

For Player 1, the formation of reputation differs slightly from that in AG; reputation evolves not only through concession behavior but also through the arrival of information. When Player 2 is rational, learning that the opponent is weak enhances Player 1's reputation, as he subsequently refrains from making further concessions. Since an informed Player 1 immediately concedes upon learning that Player 2 is behavioral, we focus on the more interesting case in which Player 2 is rational. In this case, Player 1's reputation increases through two channels: the concessions of uninformed rational types; and the gradual rise in the probability of having learned Player 2's type over time. Accordingly, given that the game has not ended, the evolution of Player 1's reputation is

$$\begin{aligned} \mu_1(t+dt \mid \text{no drop}) &= \frac{\mu_1(t) + (1-\mu_1(t))\gamma_1 dt}{1-\widehat{\lambda}_1 dt}, \\ \implies \frac{\mu_1'(t)}{\mu_1(t)} &= \frac{(1-\mu_1(t))}{\mu_1(t)}\gamma_1 + \widehat{\lambda}_1. \end{aligned} \tag{5}$$

The arrival of information accelerates the process of reputation building for Player 1. Given that Player 1 does not concede with positive probability at time zero, the time required for him to build full reputation is

$$T_1 = \frac{1}{\widehat{\lambda}_1 - \gamma_1} \ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1-z_1)}\right). \tag{6}$$

In the Appendix, we show that T_1 is strictly decreasing in both z_1 and γ_1. A higher initial reputation or a greater intensity of information acquisition accelerates Player 1's reputation accumulation. Even as z_1 approaches zero, the reputation-building time T_1 remains finite provided that γ_1 is bounded away from zero, since information arrives with positive probability over time. Consequently, Player 1 completes the reputation-building process within a finite horizon. Because $T = \min\{T_1, T_2\}$, it follows that T is always finite for any γ_1 that is bounded away from zero. This observation will be useful in the analysis of equilibrium payoffs.

3.4 Equilibrium payoffs

We now turn to the discussion of equilibrium payoffs. Let Q_i denote the probability that a rational Player i concedes at time zero, i.e., $Q_i = F_i(0)$. For rational Player i, the expected payoff is

$$u_i = (1 - z_j)Q_j \cdot a_i + \big(1 - (1 - z_j)Q_j\big)(1 - a_j).$$

Properties 2 and 3 imply that the player who requires strictly more time to build a reputation must concede with positive probability at time zero. We first focus on the case $T = T_1 < T_2$, so that Player 2 concede with positive probability at time zero. Conditional on no concession having occurred up to time $t > 0$, the posterior belief $\mu_2(t)$ evolves according to Bayes' rule, accounting for the survival of rational types. Accordingly, $\mu_2(t)$ can be expressed as

$$\mu_2(t) = \frac{z_2}{1 - \widehat{F}_2(t)} = \frac{z_2}{\big(1 - (1 - z_2)Q_2\big)\exp\Big(-\int_0^t (1 - \mu_2(s))\lambda_2(s)\,ds\Big)},$$

where $\widehat{F}_2(t)$ denotes Player 2's overall concession strategy. Since all rational Player 2 types have exited by time T_1, the terminal condition $\mu_2(T_1) = 1$ yields an implicit characterization of Q_2. Specifically,

$$Q_2 = \frac{1 - z_2 \exp\left(\int_0^{T_1} [1 - \mu_2(s)]\,\lambda_2(s)\,ds\right)}{1 - z_2} > 0. \tag{7}$$

Although the right-hand side depends on $\mu_2(\cdot)$ and $\lambda_2(\cdot)$ through the integral term, the equation uniquely pins down Q_2 because both $\mu_2(t)$ and $\lambda_2(t)$ are uniquely determined along the equilibrium path by Theorem 1.

We now consider the case in which $T_1 \geq T_2$. Player 1 must concede with probability $Q_1 \geq 0$ at time zero so that the time required for his reputation to reach one equals that of Player 2. In contrast to Player 2, Player 1's reputation evolves not only through concession behavior but also through the arrival of information. Both information arrivals and concession behavior jointly contribute to Player 1's reputation over time until it reaches one at T_2. Solving the differential equation (5) under this boundary condition gives

$$\mu_1(T_2) = 1 \implies Q_1 = \frac{\gamma_1 - \widehat{\lambda}_1 e^{(\gamma_1 - \widehat{\lambda}_1)T_2} - z_1(\gamma_1 - \widehat{\lambda}_1)}{(\gamma_1 - \widehat{\lambda}_1 e^{(\gamma_1 - \widehat{\lambda}_1)T_2})(1 - z_1)} \geq 0. \tag{8}$$

When $T_1 = T_2$, we have $Q_1 = 0$, implying that neither player concedes with a positive probability at time zero.

To summarize, by comparing T_1 with T_2 and determining each player's probability of conceding at time zero, we can derive the equilibrium expected payoff for both players. This completes the equilibrium characterization.

4 Main results

4.1 Single behavioral type space

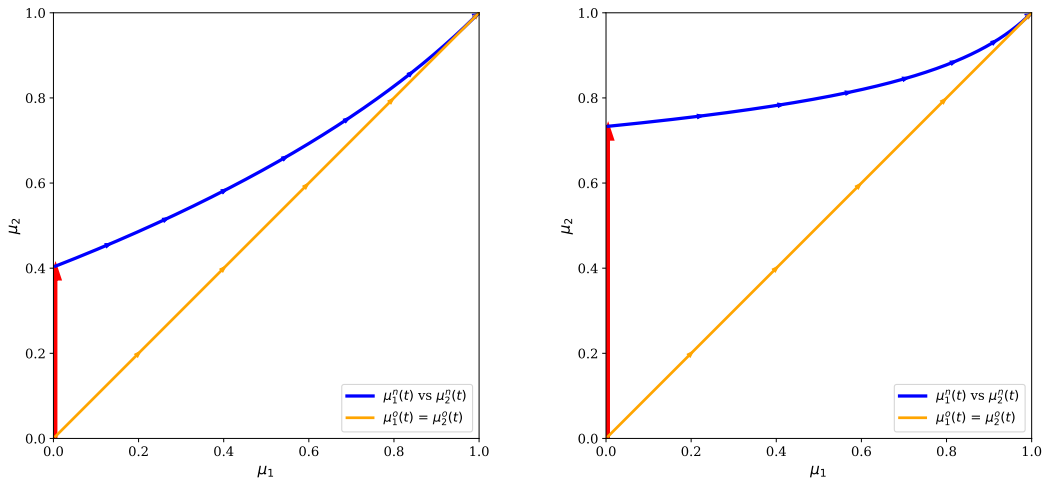
We begin by comparing the reputation dynamics in our model, where each player has a single behavioral demand, to those in AG, which provides the benchmark without information acquisition. The following proposition summarizes the key differences.

Proposition 1. *When information arrives to Player 1 at a rate of $\gamma_1 > 0$, Player 1's reputation rises strictly faster, while Player 2's rises strictly slower, relative to AG.*

In AG, each player's reputation evolves through a one−dimensional Bayesian updating process determined solely by the opponent's concession behavior. In contrast, in our environment, Player 1 additionally receives privately arriving information at rate $\gamma_1 > 0$, which alters the incentives on the equilibrium path. Once Player 1 becomes informed that the opponent is rational, he never concedes, thereby reducing the mass of uninformed rational types who would be willing to concede. To keep the uninformed rational Player 1 indifferent between conceding and waiting, Player 2 must slow down her concession behavior. These forces jointly generate a strictly faster reputation accumulation for Player 1 and a strictly slower one for Player 2 than in AG.

To illustrate the differences in reputation dynamics in a more straightforward way, Figure 2 considers the case of symmetric players. Figure 2 illustrates how unilateral information acquisition reshapes the joint reputation dynamics relative to AG. Because the arrival of information eliminates the informed Player 1's incentive to concede, Player 1 accumulates reputation more rapidly. Player 2, in turn, concedes more slowly to maintain the uninformed rational Player 1's indifference, causing her reputation to grow more gradually. Consequently, the coevolution curve $(\mu_1^n(t), \mu_2^n(t))$ is strictly flatter than its AG counterpart.

For any fixed level of Player 1's reputation, Player 2's reputation is strictly higher in our model than in AG. This difference stems from the initial concession: under the symmetric parameter values considered here, neither player concedes with positive probability at time zero in AG, whereas, with unilateral information acquisition, Player 2 concedes with a strictly positive probability at time zero. This initial upward shift in $\mu_2^n(0^+)$, together with Player 2's slower subsequent reputation growth, produces a trajectory that lies strictly above the AG benchmark for all values of μ_1.



(a) $\gamma_1 = 0.5$ (b) $\gamma_1 = 2.0$

Figure 2: Reputation coevolution under symmetric parameters $r_1 = r_2 = 0.3$, $a_1 = a_2 = 0.6$, and $z_1 = z_2 = 0.2$, for different information–arrival rates γ_1. The blue curve in each panel depicts $(\mu_1^n(t), \mu_2^n(t))$ when Player 1 can privately acquire information at rate $\gamma_1 > 0$, while the orange line depicts the benchmark coevolution $(\mu_1^o(t), \mu_2^o(t))$ in AG.

We next turn to the main results of our model and examine how the strength of information acquisition, measured by γ_1, affects equilibrium payoffs.

Proposition 2. *Let $B = (a_1, a_2, z_1, z_2, r_1, r_2, \gamma_1)$ be a bargaining game with one-sided information acquisition opportunities and single demand types, and let u_i denote Player i's equilibrium payoff in game B. Then*

$$\frac{\partial T_1}{\partial \gamma_1} < 0, \ \frac{\partial T_2}{\partial \gamma_1} > 0 \iff \frac{\partial u_1}{\partial \gamma_1} \geq 0, \ \frac{\partial u_2}{\partial \gamma_1} \leq 0.$$

For Player 2, her equilibrium expected payoff decreases as Player 1's information-acquisition intensity increases, while Player 1's expected payoff increases with γ_1. A higher information-acquisition intensity γ_1 implies that Player 1 is more likely to learn Player 2's type during the bargaining process. Consequently, Player 1's reputation builds up faster, and the time required for him to complete reputation formation, T_1, becomes shorter. At the same time, as γ_1 increases, Player 2's concession rate declines, slowing down her reputation growth and lengthening the time T_2 required for her to reach full reputation. These two effects jointly increase the probability that Player 2 concedes at time zero.

As in AG, the relative patience of the players also affects the distribution of the bargaining surplus. We focus on the case in which both players are nearly perfectly patient.

Proposition 3. *Let $\{B^n\}_n$ be a sequence of games where, for each $n \in \mathbb{N}$, $B^n = (a_1, a_2, z_1, z_2, r_1^n, r_2^n, \gamma_1)$ represents a bargaining game with one-sided information acquisition opportunities and single*

demand types. If $\lim_{n\to\infty} r_i^n = 0$ *with* $\lim_{n\to\infty} (r_1^n \ln(r_2^n))/(\ln(r_1^n)) = +\infty$*, and* u_i^n *is the equilibrium payoff for player i in the game* B^n*,*

$$\lim_{n\to\infty} \inf u_1^n \geq (1 - z_2)a_1, \qquad \lim_{n\to\infty} \sup u_2^n \leq 1 - (1 - z_2)a_1.$$

If $\lim_{n\to\infty} r_i^n = 0$ *with* $\lim_{n\to\infty} (r_2^n \ln(r_1^n))/(\ln(r_2^n)) = +\infty$*,*

$$\lim_{n\to\infty} \sup u_1^n \leq 1 - (1 - z_1)a_2, \qquad \lim_{n\to\infty} \inf u_2^n \geq (1 - z_1)a_2.$$

This result is consistent with the standard intuition in dynamic bargaining: the more patient player possesses greater bargaining power, as she incurs lower delay costs. However, compared with AG, the presence of information advantage significantly weakens the patience advantage of Player 2. Only when Player 2 is exponentially more patient than Player 1 can she compel Player 1 to make full concessions.

Consistent with prior work, we are particularly interested in the limit case of rationality, in which the prior probability that each player is behavioral is vanishingly small.

Proposition 4. *Let* $\{B^n\}_n$ *be a sequence of bargaining games with one-sided information acquisition opportunities and single demand types. If* $\lim_{n\to\infty} z_i^n = 0$ *and* u_i^n *is the equilibrium payoff for player i in the game* B^n*,*

$$\lim_{n\to\infty} u_1^n = a_1, \quad \lim_{n\to\infty} u_2^n = 1 - a_1.$$

Proposition 4 highlights how information acquisition serves as a strategic device for bargaining outcomes. When both players become nearly fully rational, Player 2 almost immediately concedes to Player 1's demand, and Player 1 obtains his proposed share in equilibrium. When Player 2's initial reputation z_2 approaches zero, if she does not concede with positive probability at time zero, her reputation–building time T_2 approaches infinity. In contrast, for Player 1, the ability to acquire information also contributes to reputation formation. According to (6), the time required for Player 1 to build full reputation, T_1, remains finite given that he does not concede with positive probability at time zero. Hence, we have $T_1 < T_2$, implying that Player 2 must concede with positive probability Q_2 at time zero. From (7), as the initial reputation z_2 is vanishingly small, the probability of immediate concession Q_2 converges to one. Therefore, in the limit, when Player 1 announces his demand at time zero, Player 2 concedes almost instantaneously.

Our findings echo the discussion of the Coase conjecture on immediate settlement in dynamic bargaining. The Coase conjecture posits that one-sided incomplete information leads to an almost instantaneous agreement that favors the informed party. When a buyer is privately informed about her valuation while the seller is uninformed, the conjecture implies that as the frequency of offers increases, the seller's commitment power vanishes, and the buyer ultimately captures

the entire surplus. AG incorporated reputational concerns into this framework by allowing both parties to be uncertain about each other's types. In their model of two-sided incomplete information, patience and the initial reputation in the limit of complete rationality jointly determine the speed of settlement and the division of surplus. Building on this insight, we show that beyond relative patience, the ability to acquire information plays an important role in the two-sided incomplete information environment. In the limit case of complete rationality, the party with the capacity to extract private information receives an immediate concession from the uninformed opponent, regardless of the spying intensity γ_1.

4.2 Rich behavioral type space

We next extend the analysis of the fully rational limit to settings with richer behavioral type spaces, where each player may be committed to a continuum of possible demands. This framework allows us to identify the behavioral types that rational players most frequently mimic and to characterize the associated equilibrium payoffs.

We first consider the case in which each player may be committed to multiple behavioral demands. Player 1 first announces his demand $a_1 \in A_1$, where A_1 denotes the set of Player 1's feasible behavioral demands. Upon observing this, Player 2 either accepts the offer $1 - a_1$ or rejects it and announces her own demand $a_2 \in A_2$, where A_2 is the set of Player 2's available behavioral demand. Let π_i denote the prior conditional probability distribution over the behavioral demands of player i, where $\pi_i(a_i)$ specifies the probability that a behavioral type demands a_i. These distributions are common knowledge. The rest of the game then proceeds as in the baseline model with one-sided information acquisition and single-type demands. Hence, a game with multiple behavioral demands and one-sided information acquisition can be represented by $B = (A_1, A_2, \pi_1, \pi_2, z_1, z_2, r_1, r_2, \gamma_1)$.

In addition to choosing when to concede, a rational player also decides which behavioral demand to mimic at the outset of the game. Let $\sigma_1 \in \Delta(A_1)$ denote the mimicking strategy of a rational Player 1 over the set of behavioral demands A_1, and $\sigma_2(\cdot \mid a_1)$ denote the strategy of a rational Player 2 conditional on observing Player 1's announced demand a_1. The action space for Player 2 includes any $a_2 \in A_2$ or Q_2, where Q_2 represents acceptance of Player 1's demand a_1.

Theorem 2. *For any bargaining game with one-sided information acquisition opportunities and multiple behavioral demand types for both players, all equilibria induce the same distribution over outcomes.*

Theorem 2 establishes the equilibrium characterization in the case with multiple behavioral demand types. The proof follows techniques similar to those in AG. Given the uniqueness of equilibrium outcomes, we next examine which behavioral type a rational Player 1 optimally chooses

to mimic when the set of behavioral demands is rich, and how Player 2 responds with her corresponding demand choice.

Fix an integer $K \geq 2$ and consider the demand set $A^K = \{2/K, 3/K, \ldots, (K-1)/K\}$. Each element of A^K represents a possible behavioral demand. For $i \in \{1, 2\}$ assume the conditional prior over behavioral demands $\pi_i \in \Delta(A^K)$ has full support on A^K. That is, every demand in A^K occurs with positive probability conditional on being a behavioral type. Let z_i^n denote the prior probability that player i is a behavioral type. Then the joint probability that player i is behavioral and conditional on being behavioral demands the share k/K equals $z_i^n\, \pi_i(k/K)$ for $k = 2, \ldots, K-1$.

Assume that both players' initial reputations vanish, without requiring them to converge at the same order. Fix K and examine the equilibrium outcomes of the bargaining game.

Proposition 5. *Let $\{B^n\}_n$ be a sequence of bargaining games with one-sided information acquisition opportunities and rich behavioral type spaces. If $\lim_{n\to\infty} z_i^n = 0$,and u_i^n is the equilibrium payoff for player i in the game B^n,*

$$\lim_{n\to\infty} \inf u_1^n \geq \frac{K-1}{K}, \quad \lim_{n\to\infty} \sup u_2^n \leq \frac{1}{K}.$$

Proposition 5 demonstrates the impact of one-sided information acquisition on surplus allocation in the fully rational limit. When there is only a single behavioral demand, Player 1 immediately receives a concession from Player 2. Intuitively, in the equilibrium with rich behavioral type spaces, Player 1 chooses to mimic the greediest demand type, while Player 2 concedes immediately to Player 1's proposal. This outcome is independent of the relative order of their initial reputations or the intensity of information acquisition γ_1. As $K \to \infty$, Player 1 effectively captures the entire surplus.

Information acquisition is efficient: it accelerates agreement and eliminates wasteful delay. Because settlement occurs almost instantaneously, Player 1's informational advantage enables immediate concessions from Player 2 even before any information is actually realized, granting Player 1 a dominant position in the negotiation.

4.3 Endogenous information acquisition

In the baseline environment, Player 1 could acquire information about Player 2's type at zero cost, and the arrival rate of information was therefore determined exogenously. In this subsection, we extend the analysis by allowing Player 1 to choose the strength of his information-acquisition technology, denoted by $\gamma_1 \geq 0$, prior to the realization of player types. This choice represents an ex ante investment in information-gathering capacity and is made before Player 1 learns whether he is rational or behavioral.

After γ_1 is chosen, Player 2 observes Player 1's information precision. Subsequently, both players' types are realized. Conditional on their types, rational players choose their demands

and specify their concession strategies. Once bargaining begins, information arrives according to a Poisson process with rate γ_1. The resulting strategy profile (σ_1, σ_2) therefore consists of Player 1's ex-ante information-acquisition decision and his continuation strategy, $\sigma_1 = (\gamma_1, F_1)$, together with Player 2's concession strategy F_2.

We consider the case in which each player has a single behavioral type. Information acquisition is technologically costly, and Player 1 incurs a strictly convex cost $c\gamma_1^2$, where $c > 0$ captures the cost of information acquisition. His expected payoff is therefore

$$u_1(\gamma_1, z_2) = (1 - z_2)Q_2(\gamma_1)a_1 + \left[1 - (1 - z_2)Q_2(\gamma_1)\right](1 - a_2) - z_1 z_2 e^{-r_1 T_1}(1 - a_2) - c\gamma_1^2,$$

where $Q_2(\gamma_1)$ is the equilibrium probability that a rational Player 2 concedes immediately at time zero. The term $(1 - z_2)Q_2(\gamma_1)$ represents the probability that Player 2 is rational and optimally chooses immediate concession. This probability increases when Player 1 appears more likely to be the commitment type or informed. Since a larger value of γ_1 accelerates the updating of Player 2's belief about Player 1's commitment power, it improves his reputational position at the outset of bargaining and increases Q_2. The term $z_1 z_2 e^{-r_1 T_1}(1 - a_2)$ represents the loss that arises when both players are behavioral since the spying decision is made before Player 1 learns his own type. In this case, settlement is unattainable and Player 1 forgoes the continuation payoff $(1 - a_2)$ that would otherwise obtain along the rational path. Proposition 2 implies that a higher spying intensity γ_1 accelerates Player 1's reputational buildup and thereby reduces T_1. As T_1 shrinks, the discounted loss in the behavioral–behavioral state becomes more severe when γ_1 rises. Player 1's continuation value therefore reflects a trade-off. Faster information arrival strengthens his reputation, but it also intensifies the loss along the behavioral–behavioral path, while the technological cost of information increases at an increasing rate.

Assuming fixed preferences and demand spaces, our goal is to examine how Player 2's initial reputation shapes Player 1's choice of information-acquisition intensity. The marginal benefit of raising γ_1 depends critically on Player 2's initial reputation z_2, which determines how informative the signals are about the likelihood that Player 2 is rational. In turn, this governs the extent to which faster information arrival can shift her beliefs and affect her willingness to concede at time zero, thereby determining the incremental value of information.

Figure 3 plots the optimal information-acquisition intensity γ_1^* as a function of z_2. When Player 2 starts with a relatively low reputation, an increase in z_2 raises the return to accelerating reputation building. In this region, Player 1 optimally chooses a positive γ_1, which increases the probability that a rational Player 2 concedes at time zero and improves his expected payoff. When Player 2's initial reputation is high, however, the incentive to accelerate reputation building becomes negligible. A high z_2 implies that Player 2 is unlikely to be rational, and thus the potential

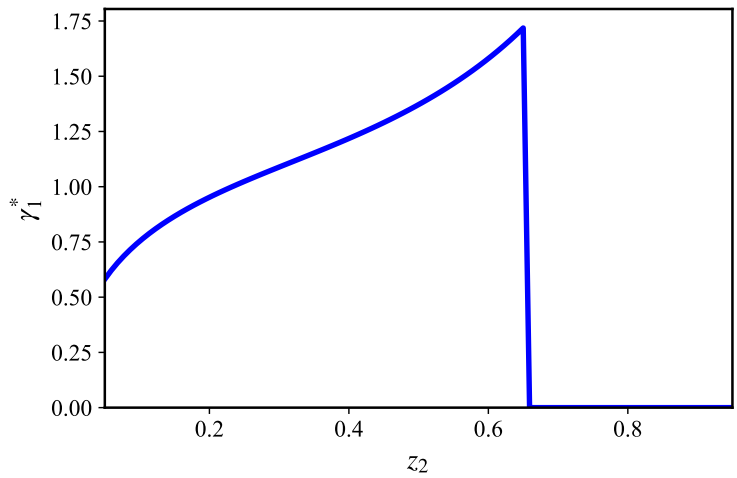
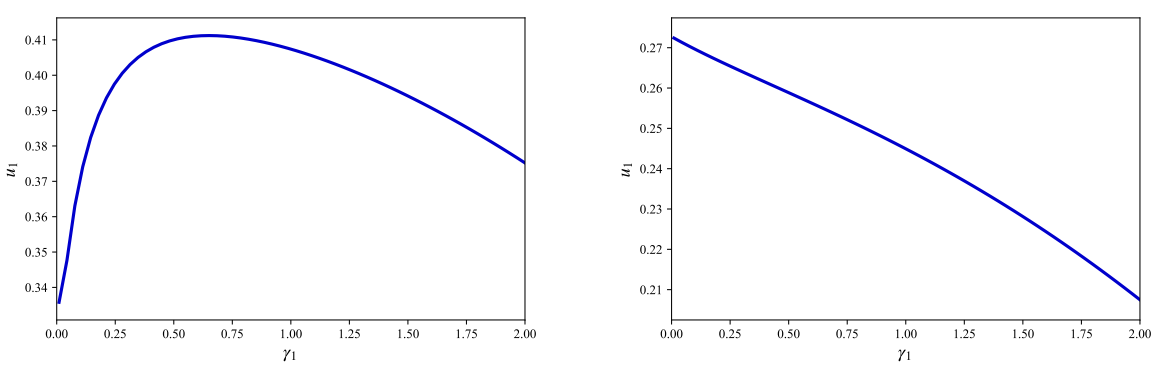


Figure 3: Optimal information-acquisition intensity γ_1^* as a function of Player 2's initial reputation z_2. Parameters are $z_1 = 0.2$, $c = 0.01$, $a_1 = a_2 = 0.6$, and $r_1 = r_2 = 0.3$.

gain from inducing a rational opponent to concede is small. In this case, the benefit of information acquisition is outweighed by its cost, and Player 1 optimally sets $\gamma_1^* = 0$. When early concessions are less likely to be induced, it is optimal for Player 1 to avoid the information cost and take the immediate payoff $1 - a_2$.



(a) $z_2 = 0.5$ (b) $z_2 = 0.9$

Figure 4: Player 1's payoff $u_1(\gamma_1, z_2)$ as a function of information-acquisition intensity. Parameters are $z_1 = 0.5$, $c = 0.01$, $a_1 = a_2 = 0.6$, and $r_1 = r_2 = 0.3$. Panel (a) considers a low initial reputation $z_2 = 0.5$. Panel (b) considers a high reputation $z_2 = 0.9$.

Figure 4 illustrates how Player 1's expected payoff varies with the information-acquisition intensity under low and high levels of Player 2's initial reputation. Panel (a) considers the case $z_2 = 0.5$, where Player 2 starts with a relatively low reputation. In this environment, u_1 initially increases in γ_1, as the reputational gains from faster belief updating dominate the technological cost of acquiring information. When the marginal cost of information is modest, a higher ac-

quisition intensity raises the likelihood that a rational Player 2 concedes immediately. Once γ_1 becomes sufficiently large, however, the convex cost dominates the reputational benefit, and u_1 begins to decline, yielding a well-defined interior optimum.

Panel (b) considers the case $z_2 = 0.9$, where Player 2 starts with a high initial reputation. In this case, u_1 is strictly decreasing in γ_1, because faster signal arrival is unlikely to induce an immediate concession by a rational opponent; equivalently, the marginal benefit of increasing the acquisition intensity is negligible relative to its cost. Consequently, the payoff is maximized at $\gamma_1 = 0$, and Player 1 optimally refrains from acquiring information.

Recalling Proposition 4, in the limiting environment in which both players are fully rational, information acquisition remains valuable even when it is endogenously chosen and entails a technological cost. In particular, if the minimal feasible cost of acquiring information exceeds the disagreement surplus $d \equiv a_1 + a_2 - 1$, then Player 1 strictly prefers to invest in information acquisition, and bargaining concludes immediately. This result shows that, in the fully rational benchmark, even costly information acquisition can almost completely eliminate delay and implement instantaneous agreement.

Taken together, these results show that information acquisition plays a nuanced and context-dependent role in bargaining. Information is valuable precisely when the opponent's initial reputation is not too high, in which case the endogenous choice of γ_1 allows Player 1 to accelerate reputation formation and improve his bargaining payoff. When the opponent already enjoys a strong reputation, by contrast, the value of information is limited and the optimal acquisition intensity is zero.

4.4 Bilateral information acquisition

We begin by analyzing reputational bargaining under *bilateral information acquisition* in the environment with a single behavioral demand for each player. We first focus on the benchmark case in which both players' information-arrival rates are exogenously given. As in the one-sided spying case, we denote by (F_1, F_2) the concession strategies of uninformed rational players, that is, players who have not yet learned the opponent's type.

Theorem 3. *Consider a bargaining game with bilateral information acquisition opportunities and single demand types, $B = (a_1, a_2, z_1, z_2, r_1, r_2, \gamma_1, \gamma_2)$, where Player i privately receives information about the opponent's type according to a Poisson process with rate γ_i. There exists a unique equilibrium. Moreover, there exists a finite time T such that every equilibrium strategy profile (F_1, F_2) satisfies the following properties:*

1. *Upon becoming informed, a rational Player i concedes immediately if he learns that Player j is behavioral, and strictly prefers to wait if he learns that Player j is rational for $t \leq T$. An*

uninformed rational player is indifferent between conceding and waiting at all times prior to becoming informed.

2. *The concession strategies F_1 and F_2 are atomless and strictly increasing on $(0, T]$, and at most one player concedes with positive probability at time $t = 0$.*

3. *Both players' reputations reach one simultaneously at time T. For all $t > T$, the only players who remain in the bargaining are either behavioral types on both sides, or rational types on both sides who are fully informed about each other's rationality. Neither player concedes with positive probability at any instant; instead, the game degenerates into a war of attrition in which informed rational players are indifferent between conceding and waiting.*

Proof. The arguments for Properties 1 and 2 are analogous to those in the proof of Theorem 1. We therefore focus on Property 3 here. Consider any time $t \leq T$. A rational and informed player who learns that the opponent is behavioral concedes immediately in order to avoid further delay costs. By contrast, upon learning that the opponent is rational, an informed rational player strictly prefers to delay and wait until time T. Uninformed rational players, on the other hand, are indifferent between conceding and waiting at all times prior to becoming informed and therefore exit the game weakly before time T.

Consequently, for all $t > T$, the only players who remain in the bargaining game are rational players who have learned that their opponents are also rational. The continuation game therefore reduces to a complete-information war of attrition. As shown by Levin (2004), such a game admits only two classes of equilibria: a degenerate equilibrium, in which one player concedes with probability one at the outset and the game ends immediately, and a nondegenerate equilibrium, in which players are indifferent between conceding and waiting and hence do not concede with positive probability at any strictly positive time.

In our framework, the degenerate equilibrium cannot arise. If, in the continuation game, one player were instead to concede with probability one at T, then an uninformed rational player would strictly benefit from deviating by delaying concession until time T, anticipating immediate concession by the opponent. Therefore, only the nondegenerate equilibrium is feasible in the continuation game, and no player concedes with positive probability at any instant $t > T$. This establishes Property 3. □

Theorem 3 characterizes the equilibrium structure of reputational bargaining under bilateral information acquisition. Prior to time T, only uninformed players may concede, whereas informed rational players strictly prefer to wait. After time T, only players who are rational and have learned that their opponents are also rational remain in the bargaining. The game therefore collapses to a complete-information war of attrition, in which informed players are indifferent

between conceding and waiting and do not concede with positive probability at any instant. This characterization also pins down the boundary condition for players' continuation values at time T. Since agreement can be reached only through the opponent's concession thereafter, the continuation value of a rational player i at time T satisfies $U_i(T) = 1 - a_j$.

For an uninformed player i, the concession rate continues to coincide with (3). When player i is endowed with an information acquisition opportunity, his reputation evolves according to

$$\mu_i(t + dt \mid \text{no drop}) = \frac{\mu_i(t) + (1 - \mu_i(t))\gamma_i \, dt}{1 - (1 - \mu_i(t))\lambda_i(t) \, dt},$$

which implies

$$\mu_i'(t) = (1 - \mu_i(t))\gamma_i + (1 - \mu_i(t))\mu_i(t)\lambda_i(t). \tag{9}$$

Hence, player i's reputation increases both through concessions by uninformed rational players and through the arrival of private information. Because a player does not observe whether the opponent has acquired information about his type, player i's continuation value from waiting at time t, conditional on knowing that player j is rational, is given by

$$U_i(t) = \left(e^{-\gamma_j t}\right)\lambda_j(t) \, dt \cdot a_i + \left[1 - \left(e^{-\gamma_j t}\right)\lambda_j(t) \, dt\right] e^{-r_i dt} \cdot U_i(t + dt),$$

which implies

$$U_i'(t) = \left(r_i + e^{-\gamma_j t}\lambda_j(t)\right)U_i(t) - a_i e^{-\gamma_j t}\lambda_j(t). \tag{10}$$

The resulting system of differential equations is analytically intractable and does not admit a closed-form solution. Nevertheless, by studying limiting cases of the bilateral information acquisition environment, we can still derive some qualitative properties of equilibrium behavior. These limiting results also provide a rationale for the one-sided information acquisition framework.

Proposition 6. *Let $\{B^n\}_n$ be a sequence of bargaining games with bilateral information acquisition opportunities. If $\lim_{n\to\infty} r_i^n = 0$ with $\lim_{n\to\infty} r_1^n / r_2^n \in (0, \infty)$, the limiting payoff bounds depend on the information arrival rates. Let u_i^n denote the equilibrium payoff of player i in the game B^n. For $\gamma_1 > \gamma_2$,*

$$\lim_{n\to\infty} \inf u_1^n \geq (1 - z_2)a_1, \qquad \lim_{n\to\infty} \sup u_2^n \leq 1 - (1 - z_2)a_1.$$

For $\gamma_1 < \gamma_2$,

$$\lim_{n\to\infty} \sup u_1^n \leq 1 - (1 - z_1)a_2, \qquad \lim_{n\to\infty} \inf u_2^n \geq (1 - z_1)a_2.$$

When players approach the limit of complete patience, reputational concerns driven by de-

lay costs and behavioral types vanish. The opportunity to acquire information becomes the sole channel through which reputation can be built and sustained. A player who receives information at a strictly higher rate is more likely to learn the opponent's rationality first and can strategically delay, thereby inducing the opponent to concede. As a result, the player with the higher information arrival rate captures almost the entire bargaining surplus.[5]

We next extend the analysis to an environment with a rich behavioral type space, in which both players can endogenously choose their information acquisition intensities γ_i at a cost $c\,\gamma_i^2$. Fix an integer $K \geq 2$ and consider the finite demand set $A^K = \left\{\frac{2}{K}, \frac{3}{K}, \ldots, \frac{K-1}{K}\right\}$. Each element of A^K corresponds to a possible behavioral demand. For each $i \in \{1, 2\}$, let $\pi_i \in \Delta(A^K)$ denote the conditional prior over behavioral demands, and assume that π_i has full support on A^K. Assume that K is sufficiently large.

The timing of the game is as follows. Player 1 first chooses an information-acquisition intensity $\gamma_1 \geq 0$. After observing γ_1, Player 2 chooses $\gamma_2 \geq 0$. Player 1 then chooses a demand a_1, after which Player 2 observes a_1 and chooses an incompatible demand a_2 or concedes immediately. From Proposition 6, equilibrium payoffs in the bargaining stage depend only on the relative magnitudes of the information-acquisition rates. Fix any (γ_1, γ_2). The induced demand subgame admits a Nash equilibrium in which the player who chooses the higher information acquisition intensity strictly prefers the maximal feasible demand $a = \bar{a}$, while the opponent's demand choice is payoff-irrelevant. Equilibrium payoffs are therefore given by $\bar{a} - c\gamma_i^2$ for the player who wins the information acquisition stage and $1 - \bar{a} - c\gamma_j^2$ for the other player.

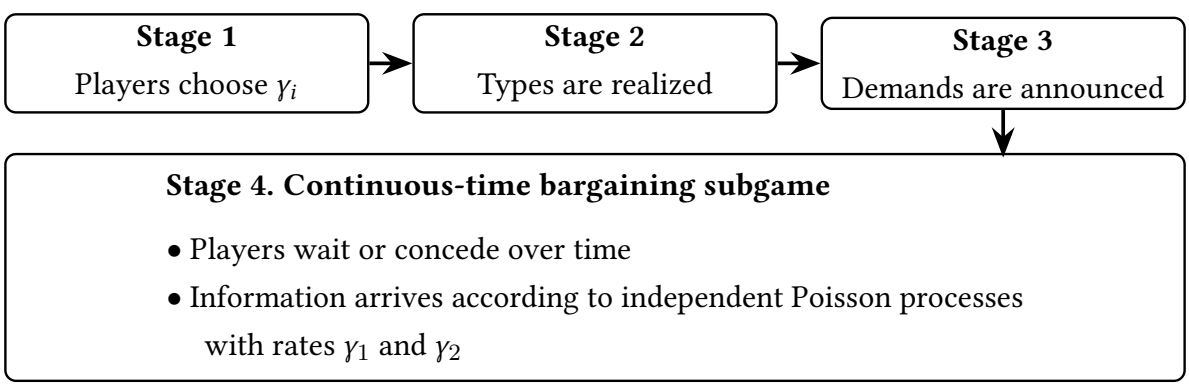


Figure 5: Timing of the bargaining game with two-sided information acquisition

Proposition 7. *Let $\{B^n\}_n$ be a sequence of bargaining games with bilateral information acquisition opportunities. Players choose information acquisition intensities sequentially, with Player 1 committing to γ_1 first and Player 2 observing γ_1 before choosing γ_2. Let $\bar{a} \in (1/2, 1)$ denote the maximal*

[5]See the proof in Appendix for the case $\gamma_1 = \gamma_2$, where concession probabilities at time zero depend on the relative magnitudes of initial reputations, demands, and patience.

feasible demand and define the net prize value $d \equiv 2\bar{a} - 1 > 0$.

When $\lim_{n\to\infty} z_i^n = 0$ *and* $\lim_{n\to\infty} r_i^n = 0$ *with* $\lim_{n\to\infty} r_1^n / r_2^n \in (0, \infty)$*, the information acquisition game admits two subgame perfect equilibria. In one equilibrium, Player 1 chooses* $\gamma_1^* = \sqrt{\frac{d}{c}}$*, Player 2 responds with* $\gamma_2^* = 0$*, and subsequently demands* $a_1 = \bar{a}$*. In the other equilibrium, Player 1 chooses* $\gamma_1^* = 0$*, Player 2 responds with* $\gamma_2^* = \sqrt{\frac{d}{c}}$*, and subsequently demands* $a_2 = \bar{a}$*. In both equilibria, the player who acquires information at the higher rate demands the maximal share, and both players obtain an equilibrium payoff equal to* $1 - \bar{a}$*.*

When players are fully rational and patient, equilibrium outcomes in the bargaining stage depend only on which player acquires information at a higher rate. Acquiring information at a strictly higher rate guarantees victory in the reputational race and yields a net prize of size $d = 2\bar{a} - 1$. As a result, the information acquisition stage becomes a sequential all-pay contest with complete information.

In such a contest, leadership determines which player captures control of the bargaining outcome but not the division of surplus. If Player 1 commits to an information acquisition intensity above the critical threshold $\sqrt{d/c}$, Player 2 optimally refrains from acquiring information, allowing Player 1 to win the reputational race uncontested. Conversely, if Player 1 commits to zero information acquisition, Player 2 optimally acquires information at rate $\sqrt{d/c}$ and becomes the effective leader in the bargaining stage. In both cases, the leader dissipates the entire net prize through information acquisition costs, equalizing equilibrium payoffs across players. Thus, endogenous information acquisition transforms reputational bargaining into a Stackelberg all-pay contest in which leadership confers control but not surplus.

It is worth noting that this conclusion extends beyond the limiting environment. Theorem 3 implies that in the equilibrium of the bilateral information acquisition model, at least one player must concede at time zero. Suppose that both players choose strictly positive information acquisition intensities, $\gamma_i > 0$ and $\gamma_j > 0$. Then, by Theorem 3, there exists a player—without loss of generality, Player i—who concedes immediately at time zero and thereby secures an expected payoff of $1 - a_j$. Since Player i anticipates conceding immediately, acquiring information yields no strategic benefit and strictly reduces his payoff. Consequently, choosing $\gamma_i > 0$ is strictly dominated by setting $\gamma_i = 0$ and conceding at time zero. More generally, whenever a player expects to concede immediately, any strictly positive information acquisition effort is payoff-irrelevant for bargaining outcomes. In such cases, the player optimally refrains from spying in order to avoid incurring unnecessary information acquisition costs.

This observation provides a rationale for the one-sided information acquisition structure adopted in the baseline model. Only a player who expects to remain in the bargaining and win the reputational race. That is, only a player who does not concede at time zero has an incentive

to acquire information with positive precision. The other player optimally sets her information acquisition intensity to zero.

5 Extensions and conclusion

In this section, we consider extensions with noisy and state-dependent information arrival to assess the robustness of our core mechanism, and then summarize the main insights of the analysis, emphasizing their relevance to practical negotiation environments.

5.1 Noisy information arrival

We begin by relaxing the assumption that information arrival perfectly reveals Player 2's type. Suppose instead that each signal is accurate with probability $1-\epsilon$, where $\epsilon \in (0, \frac{1}{2})$ denotes the misclassification probability. Upon receiving a signal, Player 1 updates his belief using Bayes' rule. A report indicating that Player 2 is rational yields

$$\mu_2^r(t) = \frac{\epsilon\, \mu_2(t)}{(1-\epsilon)(1-\mu_2(t)) + \epsilon\, \mu_2(t)} < \mu_2(t),$$

while a report indicating a behavioral type leads to

$$\mu_2^b(t) = \frac{(1-\epsilon)\mu_2(t)}{(1-\epsilon)\mu_2(t) + \epsilon(1-\mu_2(t))} > \mu_2(t).$$

Thus noisy signals attenuate belief updating by pulling posteriors toward the prior.

Proposition 8. *Let $\{B^n\}_{n\in\mathbb{N}}$ be a sequence of bargaining games with one-sided noisy information acquisition. Let u_i^n denote the equilibrium payoff of player i in B^n. If $\lim_{n\to\infty} z_i^n = 0$,*

(Single behavioral type space.) *for each n, $B^n = (a_1, a_2, z_1^n, z_2^n, r_1, r_2, \gamma_1)$ is a bargaining game with a single behavioral demand type, then*

$$\lim_{n\to\infty} \inf u_1^n = a_1, \qquad \lim_{n\to\infty} u_2^n = 1 - a_1.$$

(Rich behavioral type spaces.) *for each n, $B^n = (\pi_1, \pi_2, z_1^n, z_2^n, r_1, r_2, \gamma_1)$ is a bargaining game with rich behavioral demand spaces, where π_i has full support on $A^K = \{2/K, 3/K, \ldots, (K-1)/K\}$, then*

$$\lim_{n\to\infty} \inf u_1^n \geq \frac{(K-1)}{K}, \qquad \lim_{n\to\infty} \sup u_2^n \leq \frac{1}{K}.$$

Although signal noise lowers Player 1's continuation value after information arrival, it does not alter the strategic logic of the model. The reputation-building mechanism characterized in Sections 3 and 4 remains intact: Player 1 continues to delay strategically, rational types of Player 2 anticipate this behavior, and immediate concession persists in the fully rational limit. Hence the core comparative statics of the model are robust to imperfectly informative signals.[6]

5.2 State-dependent information arrival

We consider an alternative espionage technology in which information arrival is state-dependent. When player i incurs the spying cost, a signal arrives only if the opponent is the rational type, in which case the signal perfectly reveals rationality. If instead the opponent is the behavioral type, no signal ever arrives, even though the cost has been paid. Formally, the information technology satisfies

$$\Pr(\text{signal arrival} \mid \text{player } j \text{ is rational}) > 0,$$

and

$$\Pr(\text{signal arrival} \mid \text{player } j \text{ is behavioral}) = 0.$$

This information structure is consistent with the H-focus information environments studied in (Feinberg and Skrzypacz, 2005; Chatterjee et al., 2025), where signals are restricted to reveal only particular payoff-relevant states. In those frameworks, information acquisition is asymmetric across types: some realizations generate verifiable evidence, while others are intrinsically non-detectable. Our formulation adopts the similar logic. Espionage does not directly reveal types; rather, it detects the presence of rational optimization. A behavioral type, who follows a fixed commitment strategy and does not engage in contingent cost–benefit calculations, generates no observable trace that can be exploited by the spying technology. By contrast, rational types engage in dynamic optimization and therefore produce state-contingent information that can, in principle, be uncovered. This interpretation admits natural applications in bargaining environments. In organizational negotiations, corporate takeovers, or geopolitical conflicts, intelligence gathering typically uncovers internal deliberations, cost–benefit analyses, or strategic planning documents, all of which arise only when the opponent engages in rational optimization, whereas intrinsically committed or ideological actors generate no comparable informational trace.

Importantly, this modification does not alter the equilibrium characterization of our baseline model. For any strictly positive time, the continuation value of an uninformed player coincides with the continuation value of a player who has learned that the opponent is behavioral and there-

[6]This robustness echoes findings in static settings: Solan and Yariv (2004) show that in 2×2 games a player who spies treats probabilistic signals as if they were deterministic.

fore concedes immediately, namely $1-a_j$. Once the rational type is either identified or eliminated through belief updating, the continuation problem reduces to that in the baseline environment. Consequently, equilibrium delay, cutoff times, and surplus division remain unchanged. Therefore, this extension therefore provides a microfoundation for the implicit state-dependent information arrival mechanism already embedded in the model: espionage selectively detects rational optimization while leaving intrinsic commitment informationally inaccessible, without affecting any equilibrium predictions.

5.3 Concluding remarks

This paper studies reputational bargaining when parties can acquire information through costly spying. In our baseline analysis, information arrival accelerates reputation building by strategic delay, thereby compress the expected delay to agreement. We show that the strategic value of information is twofold: it improves learning when the opponent is rational and simultaneously disciplines the opponent's belief dynamics. In the limit of complete rationality, the information-acquiring player captures almost the entire surplus and agreement occurs rapidly. Our framework is applicable to organizational negotiations, corporate takeovers, and geopolitical conflicts, where intelligence gathering shapes bargaining power by influencing beliefs about resolve and rationality. More broadly, the model provides a tractable foundation for studying endogenous information structures in dynamic reputation games.

Several directions remain open. We can further investigate how alternative information arrival processes, such as nonstationary arrival rates, shape reputational dynamics under espionage. In particular, one may allow parties to endogenously start or terminate spying over time, thereby making information acquisition a dynamic control rather than a static choice. Beyond reputational bargaining, the framework naturally extends to dynamic games in which agents can acquire costly information about payoff-relevant states, reservation prices, or strategic constraints of their opponents. In such environments, information acquisition reshapes belief dynamics, delay rules and surplus distribution. This suggests a broader research agenda on endogenous information structures in dynamic strategic interaction.

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A Proofs

Proof of Theorem 1

Proof. Let $\sigma = (F_1, F_2)$ define a sequential equilibrium. We will argue that σ must have the form specified (i.e., uniqueness) and that these strategies do indeed define an equilibrium.

Let u_i^s denote the expected utility of a rational player i who concedes at time s. Define $A_i := \{t | u_i^t = \max_s u_i^s\}$. Since σ is an equilibrium, $A_i \neq \emptyset$ for $i = 1, 2$. Also, let $T_i = \inf\{t \geq 0 | F_i(t) = \lim_{t' \to \infty} F_i(t')\}$, where $\inf \emptyset := \infty$. Then:

(a) Rational Player 1 concedes immediately upon receiving the message that Player 2 is irrational since he knows that his opponent will never concede, and strictly prefers waiting after he receives the message that Player 2 is rational. In equilibrium, the continuation value of waiting for the informed Player 1 who knows that Player 2 is rational exceeds

that of the uninformed type. Otherwise, the uninformed Player 1 would optimally concede immediately at the outset, since doing so involves no waiting cost.

(b) $T^1 = T^2$. A rational player will not make any delay once she knows her opponent never concedes.

(c) If F_1 jumps at $t \in \mathbb{R}$, then F_2 does not jump at t. If F_1 had a jump at t, then player 2 receives a strictly higher utility by conceding an instant after t than by conceding exactly at t.

(d) If F_i is continuous at t, then u_j^s is continuous at $s = t$ for $j \neq i$.

(e) There is no interval (t', t'') such that $0 \leq t' < t'' \leq T_1$ where both F_1 and F_2 are constant on the interval (t', t''). Assume the contrary and without loss of generality, let $t^* \leq T_1$ be the supremum of t'' for which (t', t'') satisfies the above properties. Fix $t \in (t', t^*)$ and note that for ε small there exists $\delta > 0$ such that $u_i^t - \delta \geq u_i^s$ for all $s \in (t^* - \varepsilon, t^*)$ for $i = 1, 2$. By (c) and (d) there exists i such that u_i^s is continuous at $s = t^*$. Hence, for some $\eta > 0$, $u_i^s < u_i^t$ for all $s \in (t^*, t^* + \eta)$ for this player i. Since F_i is optimal, we conclude that F_i is constant on the interval $(t', t^* + \eta)$. The optimality of F_j then implies that F_j is constant on $(t', t^* + \eta)$. Hence, both functions are constant on the latter interval. This contradicts the definition of t^*.

As we noted above if F_i is constant on some interval (t', t''), then the optimality of F_j implies that F_j is constant on (t', t''); consequently, (e) implies (f):

(f) If $t' < t'' < T^1$, then $F_i(t'') > F_i(t')$ for $i = 1, 2$.

(g) F_i is continuous at $t > 0$. To see this recall that if F_i has a jump at t then F_j is constant on the interval $(t - \varepsilon, t)$ for $j \neq i$. This contradicts (f).

Given (f), it follows that A_i is dense in $[0, T_i]$. By (d), F_j is continuous on $(0, T_i)$, and hence by (d) and (g), u_i^t is continuous on $(0, T_i]$. This implies that $A_i \supseteq (0, T_i]$, which in turn implies that u_i^t is constant on $(0, T_i]$.

T_2 is finite and rational Player 2 must exit the game before T_2. To see this, suppose, conversely, rational Player 2 stays in the game with positive probability after T_2. Player 1 only concedes before and at T_2 since Player 2 makes no concession after that. Hence Player 2 will not receive any settlement from Player 1 after T_2, which is strictly dominated by conceding at T_2. Therefore, the probability of irrationality for Player 2 must reach 1 at T_2 and the continuation value of informed Player 1 is a_1. Informed Player 1 will not survive in the game after T_2. This completes our proof of the existence of the equilibrium.

We now characterize the equilibrium strategies. The evolution of Player 1's continuation value $U_1(t)$, Player 2's concession hazard $\lambda_2(t)$, and Player 2's reputation $\mu_2(t)$ is governed by the system

$$\begin{aligned}
U_1'(t) &= \big(\lambda_2(t) + r_1\big)U_1(t) - \lambda_2(t)a_1, \\
\lambda_2(t) &= \frac{r_1(1-a_2) - \gamma\big(1-\mu_2(t)\big)\big(U_1(t) - (1-a_2)\big)}{(1-\mu_2(t))(a_1+a_2-1)}, \\
\lambda_2(t) &= \frac{\mu_2'(t)}{\mu_2(t)(1-\mu_2(t))}.
\end{aligned}$$

Let $d = a_1 + a_2 - 1 > 0$. The second equation can be rewritten as

$$\lambda_2(t) = \frac{r_1(1-a_2)}{d(1-\mu_2(t))} - \frac{\gamma_1\big(U_1(t) - (1-a_2)\big)}{d}.$$

Rearranging yields

$$1 - \mu_2(t) = \frac{r_1(1-a_2)}{d\lambda_2(t) + \gamma_1 U_1(t) - \gamma(1-a_2)}. \tag{11}$$

Define

$$k(t) = 1 - \mu_2(t),$$

and

$$W(t) = \gamma_1 U_1(t) + d\lambda_2(t) - \gamma_1(1-a_2).$$

Equation (11) implies $k(t) = \frac{r_1(1-a_2)}{W(t)}$, and hence

$$k'(t) = -r_1(1-a_2)\frac{W'(t)}{W(t)^2}.$$

On the other hand, using the law of motion for $\mu_2(t)$,

$$k'(t) = -\lambda_2(t)k(t)(1-k(t)).$$

Equating the two expressions for $k'(t)$ yields

$$W'(t) = \lambda_2(t)\big(W(t) - r_1(1-a_2)\big). \tag{12}$$

Differentiating the definition of $W(t)$ gives

$$W'(t) = \gamma_1 U_1'(t) + d\lambda_2'(t).$$

Substituting for $U_1'(t)$ from the continuation-value equation and combining with (12), we obtain the reduced system

$$\begin{cases} U_1'(t) = (\lambda_2 + r_1)U_1(t) - \lambda_2 a_1, \\ \lambda_2'(t) = \lambda_2^2 + \lambda_2\left(\gamma_1 - \dfrac{r_1(1-a_2)}{d}\right) - \dfrac{\gamma_1 r_1}{d}U_1(t). \end{cases}$$

To decouple the system, define the ratio $\eta(t) = \frac{\lambda_2(t)}{U_1(t)}$. Substituting $\lambda_2(t) = \eta(t)U_1(t)$ into the second equation yields the Riccati equation

$$\eta'(t) = a_1\eta^2(t) + B\eta(t) - \frac{\gamma_1 r_1}{d}, \tag{13}$$

where $B = \gamma_1 - r_1 - \frac{r_1(1-a_2)}{d}$. The associated quadratic equation admits two real roots since

$$\Delta = B^2 + \frac{4a_1\gamma_1 r_1}{d} > 0.$$

Let η^* denote the unique positive root,

$$\eta^* = \frac{-B + \sqrt{B^2 + \frac{4a_1\gamma_1 r_1}{d}}}{2a_1} = \frac{r_1}{d} > 0.$$

Using the transformation $\eta(t) = \eta^* + v(t)^{-1}$, equation (13) can be rewritten as a linear ordinary differential equation for $v(t)$,

$$v'(t) + Kv(t) = -a_1,$$

where $K = 2a_1\eta^* + B = \gamma_1 + \frac{r_1 a_1}{d} > 0$. Solving this equation yields

$$v(t) = -\frac{a_1}{K} + C_1 e^{-Kt},$$

where C_1 is an integration constant.

Given $\eta(t)$, the continuation value $U_1(t)$ satisfies

$$U_1'(t) = U_1(t)\big(\eta(t)U_1(t) + r_1 - a_1\eta(t)\big). \tag{14}$$

Letting $w(t) = U_1(t)^{-1}$ transforms (14) into the linear equation

$$w'(t) + (r_1 - a_1\eta(t))w(t) = -\eta(t).$$

The solution is given by

$$w(t) = e^{-\int_0^t P(s)\,ds}\left(-\int_0^t \eta(\tau)e^{\int_0^\tau P(s)\,ds}\,d\tau + C_2\right), \tag{15}$$

where $P(t) = r_1 - a_1\eta(t)$ and C_2 is an integration constant. Consequently,

$$\lambda_2(t) = \eta(t)U_1(t) = \frac{\eta(t)}{w(t)}.$$

This completes the characterization of the equilibrium concession path $\lambda_2(t)$ and Player 1's continuation value $U_1(t)$.

It remains to pin down the constants C_1 and C_2, and to establish the uniqueness of Player 2's equilibrium concession time T_2. From the characterization of Player 2's concession rate, we have

$$\lambda_2(t) = \frac{\eta(t)}{w(t)} = \frac{\eta^* + \dfrac{K}{C_1Ke^{-Kt} - a_1}}{e^{-\int_0^t P(s)\,ds}\left(-\int_0^t \eta(s)e^{\int_0^s P(u)\,du}\,ds + C_2\right)},$$

where C_1 and C_2 are integration constants. Evaluating at $t = 0$ yields

$$\lambda_2(0) = \frac{\eta(0)}{w(0)} = \frac{\eta(0)}{C_2}. \tag{16}$$

The solution to the differential equation governing $\eta(t)$ is given by

$$\eta(t) = \frac{r_1}{d} + \frac{1}{-\frac{a_1}{K} + C_1e^{-Kt}},$$

which implies

$$\eta(0) = \frac{r_1}{d} + \frac{K}{KC_1 - a_1}. \tag{17}$$

Using the identities $\mu_2(0) = z_2$ and $U_1(0) = 1/C_2$ also yields

$$\lambda_2(0) = \frac{r_1(1-a_2) - \gamma_1(1-z_2)\left(\frac{1}{C_2} - (1-a_2)\right)}{(1-z_2)(a_1+a_2-1)}. \tag{18}$$

Substituting (17) into (16) and equating with (18) gives

$$\frac{1}{C_2}\left(\frac{r_1}{d} + \frac{K}{KC_1 - a_1}\right) = \frac{r_1(1-a_2) - \gamma_1(1-z_2)\left(\frac{1}{C_2} - (1-a_2)\right)}{(1-z_2)(a_1+a_2-1)}.$$

Multiplying both sides by $(1-z_2)dC_2$ and collecting terms yields

$$(1-z_2)\left(r_1+\frac{dK}{KC_1-a_1}\right)=C_2r_1(1-a_2)-\gamma_1(1-z_2)+\gamma_1(1-z_2)C_2(1-a_2).$$

Rearranging terms then gives

$$\frac{dK}{KC_1-a_1}=\frac{(1-a_2)(\gamma_1+r_1)C_2-(\gamma_1+r_1)}{1-z_2}.$$

Taking reciprocals of both sides and using $\frac{a_1}{K}=\frac{a_1d}{\gamma_1d+a_1r_1}$, we obtain

$$C_1-\frac{a_1d}{\gamma_1d+a_1r_1}=\frac{d}{(a_1-d)\left(\gamma_1+\frac{r_1}{1-z_2}\right)C_2-(\gamma_1+r_1)}, \tag{19}$$

which gives the relationship between C_1 and C_2.

Equilibrium requires that rational Player 2 concedes with probability one at time T_2, i.e., $\lim_{t\to T_2}\mu_2(t)=1$. By definition, $\lambda_2(t)$ diverges to infinity as $t\to T_2$. Hence the term

$$\eta(T_2)=\eta^*+\frac{1}{-\frac{a_1}{K}+C_1e^{-KT_2}}$$

must diverge, which yields

$$-\frac{a_1}{K}+C_1e^{-KT_2}=0,$$

and therefore $C_1=\frac{a_1}{K}e^{KT_2}$.

Moreover, Player 1's equilibrium boundary condition at the terminal time T_2 requires $U_1(T_2)=1-a_2$. Recalling that $w(t)=U_1(t)^{-1}$, evaluating (15) at $t=T_2$ yields

$$\frac{1}{1-a_2}=e^{-\int_0^{T_2}P(s)\,ds}\left(-\int_0^{T_2}\eta(\tau)e^{\int_0^s P(s)\,ds}\,dt+C_2\right).$$

Define

$$I(t)=\int_0^t P(s)\,ds=\int_0^t\Big(r_1-a_1\eta(s)\Big)ds.$$

Let $A=-\frac{a_1}{K}$ and denote $u(t)=A+C_1e^{-Kt}$, so that

$$\eta(t)=\eta^*+\frac{1}{u(t)},$$

and

$$P(t) = r_1 - a_1\eta^* - \frac{a_1}{u(t)}.$$

Define $M = r_1 - a_1\eta^* = -\frac{r_1(1-a_2)}{d} < 0$, we obtain

$$I(t) = \int_0^t \left(M - \frac{a_1}{u(s)}\right) ds = Mt - a_1 \int_0^t \frac{1}{u(s)}\, ds.$$

Using the identity $AK = -a_1$, the integral evaluates to

$$\int_0^t \frac{1}{u(s)}\, ds = -\frac{1}{a_1} \ln\left(\frac{Ae^{Kt} + C_1}{A + C_1}\right),$$

and hence

$$I(t) = Mt + \ln\left(\frac{Ae^{Kt} + C_1}{A + C_1}\right),$$

and

$$e^{I(t)} = e^{Mt}\frac{Ae^{Kt} + C_1}{A + C_1}.$$

We next compute $\int_0^{T_2} \eta(s)e^{I(t)}\, dt$. Writing

$$\eta(t)e^{I(t)} = \eta^* e^{I(t)} + \frac{1}{u(t)}e^{I(t)},$$

and noting that

$$\frac{1}{u(t)}e^{I(t)} = \frac{e^{(M+K)t}}{A + C_1},$$

we obtain

$$\int_0^{T_2} \eta(t)e^{I(t)}\, dt = \eta^* \int_0^{T_2} e^{I(t)}\, dt + \frac{1}{A + C_1}\int_0^{T_2} e^{(M+K)t}\, dt.$$

Direct calculation yields

$$\int_0^{T_2} e^{(M+K)t}\, dt = \frac{e^{(M+K)T_2} - 1}{M + K},$$

and

$$\int_0^{T_2} e^{I(t)}\, dt = \frac{1}{A + C_1}\left[\frac{A}{M + K}(e^{(M+K)T_2} - 1) + \frac{C_1}{M}(e^{MT_2} - 1)\right].$$

Substituting and using $\eta^* A + 1 = \frac{\gamma_1}{K}$, we obtain

$$\int_0^{T_2} \eta(t)e^{I(t)}\, dt = \frac{1}{A + C_1}\left[\frac{\gamma_1}{K(M + K)}(e^{(M+K)T_2} - 1) + \eta^*\frac{C_1}{M}(e^{MT_2} - 1)\right].$$

Substituting this expression into the main equation and using

$$e^{-I(T_2)} = e^{-MT_2}\frac{A + C_1}{Ae^{KT_2} + C_1},$$

we obtain

$$(Ae^{KT_2} + C_1)\frac{e^{MT_2}}{1 - a_2} = (A + C_1)C_2 - \frac{\gamma_1}{K(M + K)}(e^{(M+K)T_2} - 1) - \eta^*\frac{C_1}{M}(e^{MT_2} - 1).$$

Solving for C_2 yields

$$C_2 = \frac{1}{A + C_1}\left[(Ae^{KT_2} + C_1)\frac{e^{MT_2}}{1 - a_2} + \frac{\gamma_1}{K(M + K)}(e^{(M+K)T_2} - 1) + \eta^*\frac{C_1}{M}(e^{MT_2} - 1)\right].$$

Using the relations $C_1 = \frac{a_1}{K}e^{KT_2}$ and $A = -\frac{a_1}{K}$, we have $Ae^{KT_2}+C_1 = 0$ and $A+C_1 = \frac{a_1}{K}(e^{KT_2}-1)$. Substituting and simplifying gives

$$C_2 = \frac{\gamma_1}{a_1(M + K)}\frac{e^{(M+K)T_2} - 1}{e^{KT_2} - 1} + \frac{\eta^*}{M}\frac{e^{KT_2}(e^{MT_2} - 1)}{e^{KT_2} - 1}.$$

We now impose the consistency condition in (19)

$$C_1 - \frac{a_1 d}{\gamma_1 d + a_1 r_1} = \frac{d}{(a_1 - d)\left(\gamma_1 + \frac{r_1}{1-z_2}\right)C_2 - (\gamma_1 + r_1)}.$$

Since $\gamma_1 d + a_1 r_1 = dK$, the left-hand side reduces to $\frac{a_1}{K}(e^{K\tau_2} - 1)$. Taking reciprocals and rearranging yields

$$(1 - a_2)\left(\gamma_1 + \frac{r_1}{1 - z_2}\right)C_2 = \gamma_1 + r_1 + \frac{dK}{a_1(e^{KT_2} - 1)}.$$

Substituting the expression for C_2 and using $M + K = \gamma_1 + r_1$ together with $\frac{\eta^*}{M} = -\frac{1}{1-a_2}$, we obtain after simplification

$$\frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1}e^{(\gamma_1+r_1)T_2} - \frac{a_1 r_1 z_2}{dK}e^{KT_2} - \left(\frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1} - \frac{a_1 r_1 z_2}{dK} - (1 - z_2)\right) = 0. \tag{20}$$

Equation (20) is a scalar transcendental equation in the single unknown T_2, with all coefficients determined by the primitives of the model. We now establish that it admits a unique positive solution T_2.

Define the left-hand side of (20) as a function

$$\phi(\tau) = \frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1}e^{(\gamma_1+r_1)\tau} - \frac{a_1r_1z_2}{dK}e^{K\tau} - \left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1r_1z_2}{dK} - (1-z_2)\right),$$

Evaluating at $\tau = 0$ yields $\phi(0) = 1 - z_2 > 0$. Moreover, since $K = \gamma_1 + \frac{r_1a_1}{d} > \gamma_1 + r_1$, we have $\lim_{\tau\to\infty}\phi(\tau) = -\infty$. By continuity, there exists at least one solution $T_2 > 0$ to $\phi(T_2) = 0$. We then show that the equation $\phi(\tau) = 0$ admits a unique solution on $(0, \infty]$.

Differentiating yields

$$\phi'(\tau) = \Big(\gamma_1(1-z_2)+r_1\Big)e^{(\gamma_1+r_1)\tau} - \frac{a_1r_1z_2}{d}e^{K\tau}.$$

Define

$$R = \frac{a_1r_1z_2}{d\big(\gamma_1(1-z_2)+r_1\big)}.$$

R is strictly increasing in z_2, and there exists a unique $z_0 \in (0,1)$ given by $z_0 = \frac{d(\gamma_1+r_1)}{a_1r_1+d\gamma_1}$ such that $R = 1$.

If $z_2 \geq z_0$, then $R \geq 1$ and

$$\phi'(0) = \gamma_1(1-z_2)+r_1 - \frac{a_1r_1z_2}{d} \leq 0.$$

Moreover, for any $\tau > 0$, the inequality $\gamma_1 + r_1 < K$ implies $e^{(\gamma_1+r_1-K)\tau} < 1 \leq R$, which yields

$$\Big(\gamma_1(1-z_2)+r_1\Big)e^{(\gamma_1+r_1)\tau} < \frac{a_1r_1z_2}{d}e^{K\tau}.$$

Hence $\phi'(\tau) < 0$ for all $\tau > 0$, so $\phi(\cdot)$ is strictly decreasing on $(0, \infty)$.

If instead $z_2 < z_0$, then $R < 1$ and $\phi'(0) > 0$. In this case, the equation $\phi'(\tau) = 0$ admits a unique solution

$$\tau^* = \frac{1}{\gamma_1+r_1-K}\ln R > 0,$$

where the inequality follows from $\gamma_1 + r_1 - K < 0$ and $\ln R < 0$. Differentiating once more yields

$$\phi''(\tau) = (\gamma_1+r_1)\Big(\gamma_1(1-z_2)+r_1\Big)e^{(\gamma_1+r_1)\tau} - \frac{a_1r_1z_2K}{d}e^{K\tau},$$

and at $\tau = \tau^*$,

$$\phi''(\tau^*) = \frac{a_1r_1z_2}{d}e^{K\tau^*}(\gamma_1+r_1-K) < 0,$$

so τ^* is a global maximum. Consequently, $\phi(\cdot)$ is increasing on $[0, \tau^*]$ and decreasing on $[\tau^*, \infty)$.

In both cases, since $\phi(0) > 0$ and $\phi(\tau) \to -\infty$ as $\tau \to +\infty$, the function $\phi(\cdot)$ crosses zero

exactly once on $(0, \infty]$. Therefore, the equation $\phi(\tau) = 0$ admits a unique positive solution T_2. This establishes both the existence and uniqueness of Player 2's equilibrium concession time and completes the proof. □

Proof of Proposition 2

Proof. To establish Proposition 2, we proceed in three parts. We first show that T_2 is strictly increasing in γ_1, then show that T_1 is strictly decreasing in γ_1. Combining these comparative statics with the equilibrium characterization yields the monotonicity of equilibrium payoffs with respect to γ_1.

We begin with the comparative statics of T_2. Recall that for any $\tau > 0$,

$$\phi(\tau) = \frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1}e^{(\gamma_1+r_1)\tau} - \frac{a_1r_1z_2}{dK}e^{K\tau} - \left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1r_1z_2}{dK} - (1-z_2)\right),$$

where $K = \gamma_1 + \frac{a_1r_1}{d}$. Throughout, $a_1, a_2 \in (0,1)$ and $\gamma_1 > 0$, so that $K > \gamma_1 + r_1$. T_2 is the unique positive solution to $\phi(\tau) = 0$.

We compute the partial derivative of $\phi(\tau)$ with respect to γ_1. Applying the quotient rule yields

$$\frac{d}{d\gamma_1}\left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1}\right) = -\frac{r_1z_2}{(\gamma_1+r_1)^2}.$$

Using this identity together with $\partial_{\gamma_1}K = 1$, direct differentiation gives

$$\frac{\partial\phi}{\partial\gamma_1} = -\frac{r_1z_2}{(\gamma_1+r_1)^2}\left(e^{(\gamma_1+r_1)\tau}-1\right) + \frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1}\tau e^{(\gamma_1+r_1)\tau} + \frac{a_1r_1z_2}{dK^2}\left(e^{K\tau}-1\right) - \frac{a_1r_1z_2}{dK}\tau e^{K\tau}.$$

Let

$$x = e^{(\gamma_1+r_1)\tau},$$

so that $x > 1$ for all $\tau > 0$, and note that

$$e^{K\tau} = x^{K/(\gamma_1+r_1)},$$

with $K/(\gamma_1+r_1) > 1$. Rewriting the derivative accordingly yields

$$\begin{aligned}\frac{\partial\phi}{\partial\gamma_1} &= \frac{a_1r_1z_2}{dK^2}\left(x^{K/(\gamma_1+r_1)}-1\right) - \frac{r_1z_2}{(\gamma_1+r_1)^2}(x-1) \\ &\quad + \frac{1}{\gamma_1+r_1}\left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1r_1z_2}{dK} - (1-z_2)\right)\ln x.\end{aligned}$$

Since $K/(\gamma_1 + r_1) > 1$ and $x > 1$, the convexity of the exponential function implies

$$x^{K/(\gamma_1+r_1)} - 1 > \frac{K}{\gamma_1 + r_1}(x-1).$$

Substituting this inequality yields the lower bound

$$\frac{\partial\phi}{\partial\gamma_1} > (x-1)\left(\frac{a_1 r_1 z_2}{dK(\gamma_1+r_1)} - \frac{r_1 z_2}{(\gamma_1+r_1)^2}\right) + \frac{1}{\gamma_1+r_1}\left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1 r_1 z_2}{dK} - (1-z_2)\right)\ln x.$$

The coefficient multiplying $x-1$ can be written as

$$\frac{r_1 z_2}{(\gamma_1+r_1)^2}\left(\frac{a_1(\gamma_1+r_1)}{dK} - 1\right).$$

Using the definition of K, this expression simplifies to

$$\frac{r_1 z_2}{(\gamma_1+r_1)^2} \cdot \frac{(1-a_2)\gamma_1}{d\gamma_1 + a_1 r_1},$$

which is strictly positive. Moreover, a direct calculation shows that

$$\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1 r_1 z_2}{dK} - (1-z_2) = -(\gamma_1+r_1)\left(\frac{a_1 r_1 z_2}{dK(\gamma_1+r_1)} - \frac{r_1 z_2}{(\gamma_1+r_1)^2}\right).$$

Substituting this identity into the previous inequality yields

$$\frac{\partial\phi}{\partial\gamma_1} > \left(\frac{a_1 r_1 z_2}{dK(\gamma_1+r_1)} - \frac{r_1 z_2}{(\gamma_1+r_1)^2}\right)(x - 1 - \ln x).$$

Since $x > 1$ and $x - 1 - \ln x > 0$ for all $x > 1$, it follows that

$$\frac{\partial\phi}{\partial\gamma_1} > 0 \quad \text{whenever } \phi(\tau) = 0.$$

Because T_2 is the unique positive solution to $\phi(\tau) = 0$ and, as shown in the proof of Theorem 1, satisfies $\partial_\tau\phi(T_2) < 0$, the implicit function theorem implies

$$\frac{dT_2}{d\gamma_1} = -\frac{\partial_{\gamma_1}\phi(T_2)}{\partial_\tau\phi(T_2)} > 0.$$

Thus, an increase in γ_1 strictly increases T_2.

We next show that T_1 is strictly decreasing in γ_1. Let

$$\lambda = \widehat{\lambda}_1 = \frac{r_2(1-a_1)}{a_1+a_2-1} > 0,$$

and define

$$u(\gamma_1) = \frac{\lambda}{\lambda z_1 + \gamma_1(1-z_1)}.$$

For all $\gamma_1 \geq 0$ with $\gamma_1 \neq \lambda$, the denominator is strictly positive, so $u(\gamma_1) > 0$. Recall that in (6), T_1 can be written as

$$T_1(\gamma_1) = \frac{\ln u(\gamma_1)}{\lambda - \gamma_1}.$$

Differentiating T_1 with respect to γ_1 yields

$$\frac{dT_1}{d\gamma_1} = \frac{(\lambda-\gamma_1)\frac{d}{d\gamma_1}\ln u(\gamma_1) + \ln u(\gamma_1)}{(\lambda-\gamma_1)^2}.$$

We first compute the derivative of $\ln u(\gamma_1)$. From the definition of $u(\gamma_1)$,

$$\frac{du}{d\gamma_1} = -\frac{\lambda(1-z_1)}{(\lambda z_1 + \gamma_1(1-z_1))^2}.$$

Using the identity

$$\lambda z_1 + \gamma_1(1-z_1) = \frac{\lambda}{u(\gamma_1)},$$

we obtain

$$\frac{du}{d\gamma_1} = -\frac{(1-z_1)u(\gamma_1)^2}{\lambda}.$$

Therefore,

$$\frac{d}{d\gamma_1}\ln u(\gamma_1) = -\frac{(1-z_1)u(\gamma_1)}{\lambda}.$$

Substituting this expression into the numerator of $\frac{dT_1}{d\gamma_1}$ gives

$$(\lambda-\gamma_1)\frac{d}{d\gamma_1}\ln u(\gamma_1) + \ln u(\gamma_1) = -\frac{(1-z_1)u(\gamma_1)}{\lambda}(\lambda-\gamma_1) + \ln u(\gamma_1).$$

To simplify the first term, note that the definition of $u(\gamma_1)$ implies

$$\lambda z_1 + \gamma_1(1-z_1) = \frac{\lambda}{u(\gamma_1)},$$

which can be rearranged as

$$\gamma_1 = \frac{\lambda\left(\frac{1}{u(\gamma_1)} - z_1\right)}{1 - z_1}.$$

Hence,

$$\lambda - \gamma_1 = \lambda \frac{1 - \frac{1}{u(\gamma_1)}}{1 - z_1}.$$

Substituting this expression yields

$$\frac{(1 - z_1)u(\gamma_1)}{\lambda}(\lambda - \gamma_1) = u(\gamma_1) - 1.$$

The numerator of $\frac{dT_1}{d\gamma_1}$ therefore reduces to

$$\ln u(\gamma_1) - (u(\gamma_1) - 1).$$

Consider the function

$$g(u) = \ln u - (u - 1), \qquad u > 0.$$

We have

$$g(1) = 0,$$

and

$$g'(u) = \frac{1}{u} - 1.$$

Thus $g'(u) > 0$ for $u \in (0, 1)$ and $g'(u) < 0$ for $u > 1$, so $u = 1$ is the unique global maximizer of g. It follows that

$$g(u) < 0 \quad \text{for all } u > 0 \text{ with } u \neq 1.$$

Since $u(\gamma_1) = 1$ if and only if $\gamma_1 = \lambda$, which is excluded from the domain, we obtain

$$\ln u(\gamma_1) - (u(\gamma_1) - 1) < 0 \quad \text{for all admissible } \gamma_1.$$

The denominator $(\lambda - \gamma_1)^2$ is strictly positive, and therefore

$$\frac{dT_1}{d\gamma_1} < 0 \quad \text{for all } \gamma_1 \geq 0,\ \gamma_1 \neq \lambda.$$

Although $T_1(\gamma_1)$ is not defined at $\gamma_1 = \lambda$, both the numerator and the denominator vanish at this point. By l'Hôpital's rule, $T_1(\gamma_1)$ admits a finite limit, and thus can be continuously extended. Since $\frac{dT_1}{d\gamma_1} < 0$ for all $\gamma_1 \neq \lambda$, it follows that T_1 is strictly decreasing in γ_1 on its domain.

We next show that the stopping times T_1 and T_2 are strictly decreasing in z_1 and z_2, respec-

tively. Consider first T_1. Differentiating T_1 with respect to z_1 yields

$$\frac{\partial T_1}{\partial z_1} = \frac{1}{\widehat{\lambda}_1 - \gamma_1} \frac{\partial}{\partial z_1} \ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}\right).$$

Since

$$\frac{\partial}{\partial z_1} \ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}\right) = -\frac{\widehat{\lambda}_1 - \gamma_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)},$$

we obtain

$$\frac{\partial T_1}{\partial z_1} = -\frac{1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}.$$

The denominator is strictly positive for all admissible parameters. Hence $\partial T_1/\partial z_1 < 0$, and T_1 is strictly decreasing in z_1. When $\gamma_1 = \widehat{\lambda}_1$, the continuous extension $T_1 = (1 - z_1)/\widehat{\lambda}_1$ satisfies $\partial T_1/\partial z_1 = -1/\widehat{\lambda}_1 < 0$, so the same conclusion holds.

We now turn to T_2. Recall that T_2 is the unique positive solution to $\phi(\tau) = 0$, where

$$\phi(\tau) = \frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1} e^{(\gamma_1 + r_1)\tau} - \frac{a_1 r_1 z_2}{dK} e^{K\tau} - \left(\frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1} - \frac{a_1 r_1 z_2}{dK} - (1 - z_2)\right).$$

From the existence and uniqueness argument, it is known that $\partial_\tau \phi(T_2) < 0$.

By the implicit function theorem,

$$\frac{dT_2}{dz_2} = -\left.\frac{\partial \phi/\partial z_2}{\partial \phi/\partial \tau}\right|_{\tau = T_2}.$$

Since $\partial_\tau \phi(T_2) < 0$, the sign of dT_2/dz_2 coincides with the sign of $\partial \phi/\partial z_2$ evaluated at $\tau = T_2$. Differentiating ϕ with respect to z_2 yields

$$\frac{\partial \phi}{\partial z_2} = -\frac{\gamma_1}{\gamma_1 + r_1}\left(e^{(\gamma_1 + r_1)\tau} - 1\right) - \frac{a_1 r_1}{dK}\left(e^{K\tau} - 1\right) - 1.$$

For any $\tau > 0$, both exponential terms exceed one, and all coefficients are strictly positive. It follows that $\partial \phi/\partial z_2 < 0$, and in particular $\partial \phi/\partial z_2|_{\tau = T_2} < 0$. Therefore $dT_2/dz_2 < 0$, implying that T_2 is strictly decreasing in z_2. In conclusion, for $i = 1, 2$, the stopping time T_i is strictly decreasing in the initial reputation z_i.

Having established that T_i is strictly decreasing in z_i for $i = 1, 2$, we now complete the comparative statics argument. Define player i's reputation at time $\varepsilon > 0$, with ε arbitrarily small, as z_i^ε. Equivalently, this can be interpreted as the posterior belief at time 0^+. Since at most one player concedes at time zero with strictly positive probability, player i's posterior reputation at

time 0^+ is given by

$$z_i^0 = \frac{z_i}{z_i + (1 - z_i)(1 - Q_i)},$$

where Q_i denotes the probability that rational player i concedes at time zero. It is immediate that z_i^0 is strictly increasing in Q_i.

Consider another bargaining game $B^0 = (a_1, a_2, z_1^0, z_2^0, r_1, r_2, \gamma_1)$. Let T_i^0 denote the time it takes for player i's reputation to evolve from z_i^0 to one in the bargaining game B^0. In this equilibrium, no player concedes at time zero. The stopping times must coincide, so that $T_1^0(z_1^0) = T_2^0(z_2^0)$. Because at most one player can concede at time zero with positive probability in the original game B, there must exist at least one player i such that $z_i^0 = z_i$ and hence $T_i^0 = T_i$.

We have shown above that an increase in γ_1 strictly decreases T_1 and strictly increases T_2. Maintaining the equilibrium condition $T_1^0 = T_2^0$ therefore requires an adjustment in initial posteriors. Specifically, either z_2^0 must strictly increase or z_1^0 must strictly decrease, and at least one of these adjustments must occur. Since z_i^0 is strictly increasing in Q_i, this implies that either player 2's concession probability at time zero, Q_2, must strictly increase, or player 1's concession probability, Q_1, must strictly decrease. An increase in Q_2 or a decrease in Q_1 raises player 1's expected payoff and lowers player 2's expected payoff. Hence, an increase in γ_1 benefits player 1 and harms player 2. □

Proof of Proposition 3

Proof. Extract a subsequence of B^n such that r_1^n and r_2^n converge to their respective limits r_1 and r_2. Without loss of generality, assume that the subsequence is the sequence itself.

Case 1: $r_2|\ln r_1|/|\ln r_2| \to +\infty$**.** We first prove that the concession time T_2 strictly exceeds T_1, and we characterize Player 2's initial concession behavior Q_2.

Define $\delta := \frac{a_1}{d} - 1 > 0$. Introduce the constants $D_1 := \frac{1-a_1}{a_1+a_2-1}$, and $D_4 := \frac{1-a_2}{a_1+a_2-1}$, and define $\widehat{\lambda}_1 := D_1 r_2$, and $\widehat{\lambda}_2 := D_4 r_1$. The stopping time T_1 is given by

$$T_1 = \frac{1}{\widehat{\lambda}_1 - \gamma_1} \ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}\right).$$

As $r_2 \to 0$, we have $\widehat{\lambda}_1 = D_1 r_2 \to 0$. Expanding the prefactor yields

$$\widehat{\lambda}_1 - \gamma_1 = -\gamma_1\left(1 - \frac{D_1 r_2}{\gamma_1}\right),$$

and

$$\frac{1}{\widehat{\lambda}_1 - \gamma_1} = -\frac{1}{\gamma_1}\left(1 + \frac{D_1 r_2}{\gamma_1} + O(r_2^2)\right).$$

For the logarithmic term,

$$\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)} = \frac{D_1 r_2}{\gamma_1(1 - z_1)} \cdot \frac{1}{1 + \frac{D_1 z_1}{\gamma_1(1-z_1)} r_2},$$

and hence

$$\ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}\right) = \ln r_2 + \ln\left(\frac{D_1}{\gamma_1(1 - z_1)}\right) + O(r_2).$$

Combining the two expansions gives

$$T_1 = -\frac{1}{\gamma_1}\ln r_2 - \frac{1}{\gamma_1}\ln\left(\frac{D_1}{\gamma_1(1 - z_1)}\right) + \varepsilon_1(r_2),$$

where $\lim \varepsilon_1(r_2) = 0$. Since $\ln r_2 < 0$, this can be written as

$$T_1 = \frac{1}{\gamma_1}|\ln r_2| + D_1' + \varepsilon_1(r_2),$$

for a constant D_1'. Consequently, there exists a constant $D_T > 0$ such that, for all sufficiently small r_2,

$$\frac{1}{\gamma_1}|\ln r_2| - D_T \le T_1 \le \frac{1}{\gamma_1}|\ln r_2| + D_T. \tag{21}$$

We next analyze T_2, which satisfies

$$\frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1} e^{(\gamma_1 + r_1)T_2} - \frac{a_1 r_1 z_2}{dK} e^{KT_2} - \left(\frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1} - \frac{a_1 r_1 z_2}{dK} - (1 - z_2)\right) = 0,$$

where $K = \gamma_1 + \frac{a_1}{d} r_1$. Define the following quantities:

$$\alpha := \frac{\gamma_1(1 - z_2) + r_1}{\gamma_1 + r_1},$$

$$\beta := \frac{a_1 r_1 z_2}{dK},$$

and

$$\kappa_2 := \alpha - \beta - (1 - z_2).$$

Then T_2 is characterized by the equation

$$\alpha e^{(\gamma_1+r_1)T_2} - \beta e^{KT_2} = \kappa_2. \tag{22}$$

As $r_1 \to 0$, straightforward expansions yield

$$\alpha = 1 - z_2 + \frac{r_1 z_2}{\gamma_1} + O(r_1^2),$$

and

$$\beta = \frac{a_1 z_2}{d\gamma_1} r_1 + O(r_1^2).$$

Consequently,

$$\kappa_2 = -\frac{z_2 \delta}{\gamma_1} r_1 + O(r_1^2),$$

which is strictly negative for all sufficiently small r_1. Since $\kappa_2 = O(r_1)$, it follows that the solution T_2 must diverge as $r_1 \to 0$.

Introduce the rescaled variable

$$x := r_1 \delta T_2,$$

so that

$$KT_2 = (\gamma_1 + r_1)T_2 + x.$$

Substituting this relation into (22) yields

$$e^{(\gamma_1+r_1)T_2}\left(\alpha - \beta e^x\right) = \kappa_2.$$

Because $\kappa_2 < 0$ and the exponential term is strictly positive, one must have $\alpha - \beta e^x < 0$, which implies that βe^x is asymptotically close to α. Accordingly, write

$$\beta e^x = \alpha - \xi,$$

where $\xi > 0$ is exponentially small. Taking logarithms gives

$$x = \ln\left(\frac{\alpha}{\beta}\right) + o(1).$$

A direct computation shows that

$$\frac{\alpha}{\beta} = \frac{d\gamma_1(1 - z_2)}{a_1 z_2} \cdot \frac{1}{r_1}\big(1 + O(r_1)\big).$$

Therefore,

$$x = |\ln r_1| + D_2 + o(1),$$

where

$$D_2 := \ln\left(\frac{d\gamma_1(1-z_2)}{a_1 z_2}\right).$$

Substituting back yields

$$T_2 = \frac{|\ln r_1|}{r_1\delta} + \frac{D_2}{r_1\delta} + o\left(\frac{1}{r_1}\right).$$

Hence, there exists a constant $D_T' > 0$ such that, for all sufficiently small r_1,

$$\frac{|\ln r_1|}{r_1\delta} - \frac{D_T'}{r_1} \le T_2 \le \frac{|\ln r_1|}{r_1\delta} + \frac{D_T'}{r_1}. \tag{23}$$

Comparing (21) and (23), it suffices to verify that

$$\frac{|\ln r_1|}{r_1\delta} > \frac{1}{\gamma_1}|\ln r_2| + o\left(\frac{1}{r_1}\right).$$

Under the maintained condition

$$\lim_{r_1,r_2\to 0} \frac{r_2|\ln r_1|}{|\ln r_2|} = +\infty,$$

one has

$$\frac{|\ln r_1|}{r_1|\ln r_2|} \to +\infty,$$

which implies that the leading term in T_2 dominates that of T_1. Hence Player 2 concedes with positive probability Q_2 at time zero.

We now study the limiting behavior of Q_2. From (7), we construct a lower bound by fixing a upper bound of concession rate at the constant value

$$\widehat{\lambda}_2 = \frac{r_1(1-a_2)}{a_1+a_2-1} = D_4 r_1.$$

Let

$$\widehat{Q}_2 \equiv \frac{1 - z_2\exp(\widehat{\lambda}_2 T_1)}{1-z_2}. \tag{24}$$

By construction, $\widehat{Q}_2 < Q_2 \le 1$, since replacing the time-varying hazard rate by its constant counterpart weakly decreases the required reputation-building time. Using the upper bound in

(21),

$$r_1 T_1 \leq \frac{r_1}{\gamma} |\ln r_2| + D_T r_1.$$

We now show that the imposed condition $\lim_{r_1,r_2\to 0} \frac{r_2|\ln r_1|}{|\ln r_2|} = +\infty$ implies $r_1|\ln r_2| \to 0.$, and hence $Q_2 \to 1$. Suppose, to the contrary, that this conclusion fails. Then there exist a constant $c > 0$ and a sequence $(r_1^{(n)}, r_2^{(n)}) \to (0,0)$ such that $r_1^{(n)}|\ln r_2^{(n)}| \geq c$ for all n. This implies

$$|\ln r_2^{(n)}| \geq \frac{c}{r_1^{(n)}}.$$

Substituting this bound into the ratio appearing in the imposed condition yields

$$\frac{r_2^{(n)}|\ln r_1^{(n)}|}{|\ln r_2^{(n)}|} \leq \frac{r_2^{(n)}|\ln r_1^{(n)}|}{c/r_1^{(n)}} = \frac{r_1^{(n)} r_2^{(n)}|\ln r_1^{(n)}|}{c}.$$

Since $r_1^{(n)} \to 0$, we have $r_1^{(n)}|\ln r_1^{(n)}| \to 0$, and since $r_2^{(n)} \to 0$ as well, it follows that $r_1^{(n)} r_2^{(n)}|\ln r_1^{(n)}| \to 0$. Consequently,

$$\frac{r_2^{(n)}|\ln r_1^{(n)}|}{|\ln r_2^{(n)}|} \to 0,$$

which contradicts the imposed condition. Therefore, the supposition is false, and we conclude that $r_1|\ln r_2| \to 0$. Consequently, we have $\widehat{Q}_2 \to 1$. Because Q_2 is bounded above by one and bounded below by $\widehat{Q}_2$, it follows that $Q_2 \to 1$. This result implies that Player 2 concedes almost immediately with probability arbitrarily close to one.

Case 2: $r_1|\ln r_2|/|\ln r_1| \to +\infty$**.** Recall from Case 1 we establish that

$$T_1 = \frac{1}{\gamma_1}|\ln r_2| + O(1),$$

and

$$T_2 = \frac{|\ln r_1|}{r_1 \delta} + O\left(\frac{1}{r_1}\right).$$

Since

$$\lim_{r_1,r_2\to 0} \frac{r_1|\ln r_2|}{|\ln r_1|} = +\infty,$$

equivalently, we have

$$\frac{|\ln r_2|}{|\ln r_1|/r_1} \to +\infty.$$

Since $|\ln r_1|/r_1 \to +\infty$, this condition implies that the leading term $\frac{1}{\gamma_1}|\ln r_2|$ in T_1 dominates the leading term $\frac{|\ln r_1|}{r_1\delta}$ in T_2. The lower-order remainder terms in the expressions for T_1 and T_2 are negligible by comparison. Therefore, for all (r_1, r_2) sufficiently small, we have $T_1 > T_2$. Hence Player 1 concedes with positive probability Q_1 at time zero.

We now characterize the limiting behavior of Q_1. Using the closed-form expression in (8), we can write

$$Q_1 - 1 = \frac{z_1\widehat{\lambda}_1\big(1 - e^{(\gamma_1-\widehat{\lambda}_1)T_2}\big)}{(\gamma_1 - \widehat{\lambda}_1 e^{(\gamma_1-\widehat{\lambda}_1)T_2})(1 - z_1)}.$$

Since $\widehat{\lambda}_1 = O(r_2)$ and $T_2 \to +\infty$, it follows that

$$Q_1 - 1 = -\frac{z_1}{\gamma_1(1 - z_1)}\,\widehat{\lambda}_1 e^{(\gamma_1-\widehat{\lambda}_1)T_2}\big(1 + o(1)\big).$$

Using the asymptotic expression for T_2, we obtain

$$\ln\big(\widehat{\lambda}_1 e^{(\gamma_1-\widehat{\lambda}_1)T_2}\big) = -|\ln r_2| + \frac{\gamma_1}{\delta}\frac{|\ln r_1|}{r_1} + o\left(\frac{|\ln r_1|}{r_1}\right).$$

Under the maintained condition, the right-hand side converges to $-\infty$, which implies

$$\widehat{\lambda}_1 e^{(\gamma_1-\widehat{\lambda}_1)T_2} \to 0.$$

Consequently, $Q_1 \to 1$. We therefore conclude that Player 1 concedes almost immediately with a probability arbitrarily close to one. This establishes the result. □

Proof of Proposition 4

Proof. Extract a subsequence of B^n such that z_1^n and z_2^n converge to their respective limits z_1 and z_2. Without loss of generality, assume that the subsequence is the sequence itself.

We first analyze the exit time T_1. By definition in (6),

$$T_1 = \frac{1}{\widehat{\lambda}_1 - \gamma_1}\ln\left(\frac{\widehat{\lambda}_1}{\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1)}\right).$$

When the initial belief z_1 is sufficiently small, the denominator satisfies

$$\widehat{\lambda}_1 z_1 + \gamma_1(1 - z_1) = \gamma_1 + (\widehat{\lambda}_1 - \gamma_1)z_1 \approx \gamma_1,$$

where the approximation is arbitrarily accurate for z_1 small. Consequently,

$$T_1 \approx \frac{1}{\widehat{\lambda}_1 - \gamma_1} \ln\left(\frac{\widehat{\lambda}_1}{\gamma_1}\right),$$

and hence T_1 remains bounded uniformly for all sufficiently small z_1.

We next consider the exit time T_2, defined as the unique positive solution to

$$\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} e^{(\gamma_1+r_1)T_2} - \frac{a_1 r_1 z_2}{dK} e^{KT_2} - \left(\frac{\gamma_1(1-z_2)+r_1}{\gamma_1+r_1} - \frac{a_1 r_1 z_2}{dK} - (1-z_2)\right) = 0.$$

For z_2 small, the constant term in parentheses is of order z_2, and thus negligible relative to the exponential terms. In order for the equation to hold, the two exponential terms must therefore be of comparable magnitude. This requires T_2 to be large when z_2 is small.

In this regime, the defining equation implies the asymptotic balance

$$e^{(\gamma_1+r_1)T_2} \approx z_2\, e^{KT_2},$$

where the approximation becomes arbitrarily accurate as z_2 decreases. Taking logarithms yields

$$(K - (\gamma_1 + r_1))T_2 \approx \ln\left(\frac{1}{z_2}\right).$$

Since $K - (\gamma_1 + r_1) = \frac{r_1(1-a_2)}{d} > 0$, we obtain the asymptotic representation

$$T_2 \approx \frac{d}{r_1(1-a_2)} \ln\left(\frac{1}{z_2}\right),$$

where the approximation is arbitrarily accurate for sufficiently small z_2. In particular, T_2 becomes arbitrarily large as z_2 decreases.

Combining the above results, T_1 is uniformly bounded for small z_1, whereas T_2 grows without bound as z_2 becomes small. Therefore, there exists $\varepsilon > 0$ such that $T_1 < T_2$ whenever $z_1, z_2 < \varepsilon$. Player 2 concedes with positive probability Q_2 at time zero. Consider

$$\widehat{Q}_2 = \frac{1 - z_2 e^{\widehat{\lambda}_2 T_1}}{1 - z_2}$$

in (24). Since T_1 is bounded, the factor $e^{\widehat{\lambda}_2 T_1}$ is finite. Hence, for z_2 sufficiently small,

$$z_2 e^{\widehat{\lambda}_2 T_1} \approx 0,$$

with an approximation that can be made arbitrarily accurate. It follows that $\widehat{Q}_2$. converges to one. By construction, $\widehat{Q}_2 < Q_2 \leq 1$, and therefore Q_2 is also arbitrarily close to one for sufficiently small z_2. □

Proof of Theorem 2

Before proving Theorem 2, we first prove a lemma that shows the uniqueness of equilibrium $B = (a_1, \pi_2, x, z_2, r_1, r_2, \gamma_1)$ by replacing z_1 with x and A_1 with $\{a_1\}$, i.e., Player 1 has a single demand type and Player 2 has multiple demand types.

Lemma A.1. *For any game $B = (a_1, \pi_2, x, z_2, r_1, r_2, \gamma_1)$ with information acquisition opportunities for Player 1, a single demand for Player 1, and multiple demands for Player 2, there exists a unique equilibrium.*

Proof. Denote by $\sigma_2(\cdot)$ a probability distribution over $A_2 \cup \{Q\}$, a mimicking strategy of an unjustified player 2. Since mimicking $a_2 < 1 - a_1$ is never optimal and mimicking $a_2 = 1 - a_1$ is equivalent to conceding, we assume that in equilibrium $\sigma_2(a_2) = 0$ for all $a_2 \leq 1 - a_1$. If $x = 1$, then in equilibrium $\sigma_2(Q) = 1$, because an unjustified player 2 will not delay conceding if she knows that player 1 is justified. For the remainder of the proof, we assume $x < 1$.

Define $T_i(a_1, a_2, x)$ as the time it takes for player i's reputation to increase from x to 1 on the equilibrium reputation path when each player i's demand is a_i. Note that $T_1(a_1, a_2, x)$ is continuous and strictly decreasing in x on $(0, 1)$, and $T_2(a_1, a_2, y)$ is continuous and strictly decreasing in y on $(0, 1)$.

It remains to be shown that a rational Player 2's equilibrium behavior $\sigma_2(\cdot)$ and a rational Player 1's conceding behavior $Q_1(a_1, a_2, x, \sigma_2)$ at time zero are uniquely determined. Subsequently, we provide a series of definitions and use them to prove a series of claims that lead to equilibrium existence and uniqueness. Define Player 2's reputation at time 0 when she plays a_2 with probability σ_2 as

$$y^*(a_2, \sigma_2) = \frac{z_2 \pi_2(a_2)}{z_2 \pi_2(a_2) + (1 - z_2)\sigma_2}.$$

Note that the more likely a rational player 2 is to announce a particular demand a_2, the more likely she is believed to be rational, and the lower her payoff from demanding a_2 is.

Let $\bar{\sigma}_2(a_1, a_2, x)$ be the maximum probability player 2 plays a_2 in equilibrium so that the expected payoff from demanding a_2 is higher than directly conceding to Player 1's demand. For any $a_2 < 1 - a_1$, $\bar{\sigma}_2(a_1, a_2, x) = 0$ because conceding to Player 1's demand a_1, which results in a payoff of $1 - a_1$, is a strictly better strategy than demanding strictly less than $1 - a_1$ and a weakly better strategy than demanding $1 - a_1$. For any $a_2 > 1 - a_1$, after choosing a_2, in any equilibrium, Player 2 should not concede with a positive probability at time 0. First, if Player 1's reputation can

reach 1 without conceding with a positive probability at time 0 and Player 2's reputation reaches 1 slower than Player 1 when she demands a_2 with probability 1, $\bar{\sigma}_2(a_1, a_2, x)$ is the unique solution of σ_2 to $T_1(a_1, a_2, x) = T_2(a_1, a_2, y^*(a_2, \sigma_2))$ so that the two players' reputations reach 1 at the same time.

Note that in equilibrium $\sigma_2(a_2) \leq \bar{\sigma}_2(a_1, a_2, x)$ for all $a_2 \geq 1 - a_1$. To see why this claim must hold, suppose player 2 mimics a_2 with a probability strictly higher than $\bar{\sigma}_2(a_1, a_2, x) < 1$. Then player 2 must concede with a strictly positive probability at time 0 in order for the two players' reputations to reach 1 at the same time. However, we have specified that player 2 does not concede at time zero after announcing her demand. Second, if player 1's reputation reaches 1 even slower than when player 2 demands a_2 with probability 1, $\bar{\sigma}_2(a_1, a_2, x) = 1$. This scenario happens whenever $T_1(a_1, a_2, x) > T_2(a_1, a_2, y^*(a_2, 1))$.

In summary, in any equilibrium,

$$\sigma_2(a_2) = 0, \quad \text{if } a_2 < 1 - a_1; \qquad \sigma_2(a_2, x) = \bar{\sigma}_2(a_1, a_2, x), \quad \text{if } a_2 \geq 1 - a_1$$

and

$$T_1(a_1, a_2, x) = T_2(a_1, a_2, y^*(a_2, \sigma_2(a_2))).$$

When Player 2 demands a_2 with probability σ_2 and a rational Player 1 concedes with probability $Q_1(a_1, a_2, x, \sigma_2)$, Player 1 must raise his time 0 reputation to $x^*(a_1, a_2, \sigma_2)$ so that their reputations reach 1 at the same time:

$$T_1(a_1, a_2, x^*(a_1, a_2, \sigma_2)) = T_2(a_1, a_2, y^*(a_2, \sigma_2)).$$

In order to do so, a rational Player 1 concedes with probability

$$Q_1(a_1, a_2, x, \sigma_2) = 1 - x\frac{1 - x^*(a_1, a_2, \sigma_2)}{1 - x}\frac{1}{x^*(a_1, a_2, \sigma_2)},$$

so that player 1's reputation is raised to

$$x^*(a_1, a_2, \sigma_2) = \frac{x}{x + (1 - x)[1 - Q_1(a_1, a_2, x, \sigma_2)]}.$$

When player 2 demands a_2 with probability σ_2 and an unjustified player 1 concedes with probability $Q_1(a_1, a_2, x, \sigma_2)$, an unjustified player 2's expected payoff is

$$u_2^i(a_1, a_2, x, \sigma_2) = 1 - a_1 + (1 - x)Q_1(a_1, a_2, x, \sigma_2)(a_1 + a_2 - 1).$$

Two additional properties restrict player 2's equilibrium strategy $\sigma_2(\cdot)$. First, for any a_2 and

$a_2' > a_2$, if $\sigma_2(a_2) > 0$, then $\sigma_2(a_2') > 0$. We prove this property by contradiction. Suppose $\sigma_2(a_2) > 0$ and $\sigma_2(a_2') = 0$. Because $\sigma_2(a_2') = 0$, $u_2^i(a_1, a_2', x, \sigma_2(a_2')) = 1 - a_1 + (1-x)(a_1 + a_2' - 1)$. Because $\sigma_2(a_2) > 0$, $u_2^i(a_1, a_2, x, \sigma_2(a_2)) = 1 - a_1 + (1-x)Q_1(a_1, a_2, x, \sigma_2(a_2))(a_1 + a_2 - 1) \leq 1 - a_1 + (1-x)(a_1 + a_2 - 1) = u_2^i(a_1, a_2', x, \sigma_2(a_2'))$. Second, whenever $\sum_{a_2} \bar{\sigma}_2(a_1, a_2, x) < 1$, $\sigma_2(a_2) = \bar{\sigma}_2(a_1, a_2, x)$ for all $a_2 \geq 1 - a_1$. The two properties together imply that we only need to check first if $\sum_{a_2} \bar{\sigma}_2(a_1, a_2, x) \leq 1$, and, if the first condition does not hold, then we find the equilibrium strategy among the set of strategies $\sigma_2(\cdot)$ such that $\sigma_2(a_2') > 0$ for all $a_2' \geq a_2$, for each $a_2 \in A_2$.

Denote by

$$\Delta_2(a_1, x) := \left\{ \sigma_2(\cdot) \in \Delta(A_2 \cup \{Q\}) \;\middle|\; \begin{array}{ll} \sigma_2(a_2) = 0, & \forall a_2 \leq 1 - a_1, \\ \sigma_2(a_2) = \bar{\sigma}_2(a_1, a_2, x), & \forall a_2 > 1 - a_1 \end{array} \right\}.$$

the set of candidate equilibrium mimicking strategies of Player 2 in the game $B_1(a_1, x)$, where Δ denotes the set of all probability distributions on $A_2 \cup \{Q\}$. Note that the set $\Delta_2(a_1, x)$ is nonempty, convex, and compact. For any candidate equilibrium mimicking strategy $\sigma_2(\cdot) \in \Delta_2(a_1, x)$, define

$$\widehat{u}_2(x, \sigma_2(\cdot)) := \min_{a_2 : \sigma_2(a_2) > 0} u_2^*(a_1, a_2, x, \sigma_2(a_2)).$$

Explicitly,

$$\widehat{u}_2(x, \sigma_2(\cdot)) := \begin{cases} \min_{a_2 : \sigma_2(a_2) > 0} u_2^*(a_1, a_2, x, \sigma_2(a_2)), & \text{if } \sigma_2(Q) = 0, \\ 1 - a_1, & \text{if } \sigma_2(Q) \neq 0. \end{cases}$$

Note that $\widehat{\sigma}_2(\cdot)$ is an equilibrium strategy iff $\widehat{\sigma}_2(\cdot)$ solves $\max_{\sigma_2(\cdot) \in \Delta_2(a_1, x)} \widehat{u}_2(x, \sigma_2(\cdot))$. ($\Rightarrow$) Suppose $\widehat{\sigma}_2(\cdot)$ is an equilibrium strategy. Any equilibrium strategy $\sigma_2(\cdot)$ satisfies that for all $a_2 \in A_2 \cup \{Q\}$ such that $\sigma_2(a_2) > 0$, $u_2^*(a_1, a_2, x, \sigma_2(a_2))$ is the same. If $\sigma_2(Q) > 0$, then

$$u_2^*(a_1, a_2, x, \sigma_2(a_2)) = 1 - a_1;$$

if $\sigma_2(Q) = 0$, then

$$u_2^*(a_1, a_2, x, \sigma_2(a_2)) = \min_{a_2 : \widehat{\sigma}_2(a_2) > 0} u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2)).$$

Hence, any equilibrium strategy $\widehat{\sigma}_2(\cdot)$ must generate an equilibrium utility of $\widehat{u}_2(x, \sigma_2(\cdot))$. Hence, $\widehat{\sigma}_2(\cdot)$ maximizes $\widehat{u}_2(x, \sigma_2(\cdot))$ among all candidate equilibrium strategies $\sigma_2(\cdot)$. ($\Leftarrow$) Suppose $\widehat{\sigma}_2(\cdot)$ solves $\max_{\sigma_2(\cdot) \in \Delta_2(a_1, x)} \widehat{u}_2(x, \sigma_2(\cdot))$. By the strict monotonicity of $u_2^*(a_1, a_2, x, \cdot)$, for all $a_2 \in A_2$ such that $\widehat{\sigma}_2(a_2) > 0$, $u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2)) = \widehat{u}_2(x, \widehat{\sigma}_2(\cdot))$. Coupled with the fact that $\widehat{\sigma}_2(\cdot)$

is the feasible strategy that maximizes $\widehat{u}_2(x, \sigma_2(\cdot))$, $\widehat{\sigma}_2(\cdot)$ is an equilibrium strategy.

We now define a correspondence

$$\Gamma(\tilde{\sigma}_2) : \Delta_2(a_1, x) \rightrightarrows \Delta_2(a_1, x)$$

as follows:

$$\left\{\tilde{\sigma}_2'(\cdot) \in \Delta_2(a_1, x) : \ \tilde{\sigma}_2'(a_2) > 0 \Rightarrow u_2^i(a_1, a_2, x, \tilde{\sigma}_2(a_2)) \geq u_2^i(a_1, a_2', x, \tilde{\sigma}_2(a_2')) \ \forall a_2'\right\}.$$

It follows that $\tilde{\sigma}_2$ is an equilibrium strategy iff and only if $\tilde{\sigma}_2$ is a fixed point of Γ. ($\Rightarrow$) Suppose $\widehat{\sigma}_2(\cdot)$ solves $\max_{\sigma_2(\cdot)\in\Delta_2(a_1,x)} \widehat{u}_2(x, \sigma_2(\cdot))$. By the argument above, $\widehat{\sigma}_2(\cdot)$ is an equilibrium strategy. Therefore, $\widehat{\sigma}_2(a_2) > 0$ implies $u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2)) \ \geq \ u_2^*(a_1, a_2', x, \widehat{\sigma}_2(a_2'))$ for any $a_2' \in A_2$. By the definition of Γ, $\widehat{\sigma}_2(\cdot) \in \Gamma(\widehat{\sigma}_2(\cdot))$. ($\Leftarrow$) Suppose $\widehat{\sigma}_2(\cdot) \in \Gamma(\widehat{\sigma}_2(\cdot))$. By the definition of Γ, $\widehat{\sigma}_2(a_2) > 0$ implies

$$u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2)) \ \geq \ u_2^*(a_1, a_2', x, \widehat{\sigma}_2(a_2')) \quad \text{for any } a_2' \in A_2.$$

Assume by contradiction that $\widehat{\sigma}_2(\cdot)$ does not solve $\max_{\sigma_2(\cdot)\in\Delta_2(a_1,x)} \widehat{u}_2(x, \sigma_2(\cdot))$ but $\widetilde{\sigma}_2(\cdot) \neq \widehat{\sigma}_2(\cdot)$ does. There must exist an $a_2 \in A_2$ such that $\widehat{\sigma}_2(a_2) > 0$ and $\widetilde{\sigma}_2(a_2) < \widehat{\sigma}_2(a_2)$ (otherwise, if $\widetilde{\sigma}_2(a_2) \geq \widehat{\sigma}_2(a_2)$ for all a_2 such that $\widehat{\sigma}_2(a_2) > 0$, then by the strict monotonicity of u_2^*, $u_2^*(a_1, a_2, x, \widetilde{\sigma}_2(a_2)) \leq u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2))$, and $\widehat{u}_2(x, \widetilde{\sigma}_2(\cdot)) \leq \widehat{u}_2(x, \widehat{\sigma}_2(\cdot))$). However, that implies that there exists $a_2' \in A_2 \cup \{Q\}$ such that $\widetilde{\sigma}_2(a_2') > \widehat{\sigma}_2(a_2')$. If $a_2' = Q$, then $\widehat{u}_2(x, \widetilde{\sigma}_2(\cdot)) \leq \widehat{u}_2(x, \widehat{\sigma}_2(\cdot))$. If $a_2' \in A_2$, then $\widehat{u}_2(x, \widetilde{\sigma}_2(\cdot)) \leq \widehat{u}_2(x, \widehat{\sigma}_2(\cdot))$.

Equilibrium existence follows from Kakutani's fixed-point theorem, as $\Delta_2(a_1, x)$ is nonempty, convex, and compact, and Γ is convex-valued and upper hemicontinuous. Uniqueness follows from the strict monotonicity of $u_2^i(a_1, a_2, x, \cdot)$ in x: if two equilibrium strategies assign different probabilities to some $a_2 > 1 - a_1$, strict inequality leads to a contradiction. Suppose there are two equilibrium strategies $\widehat{\sigma}_2(\cdot)$ and $\widetilde{\sigma}_2(\cdot)$; without loss of generality, suppose $\widehat{\sigma}_2(a_2) > \widetilde{\sigma}_2(a_2) > 0$ for some $a_2 > 1 - a_1$. The utilities of playing the two strategies are different:

$$\widehat{u}_2(x, \widehat{\sigma}_2(\cdot)) = u_2^*(a_1, a_2, x, \widehat{\sigma}_2(a_2)) < u_2^*(a_1, a_2, x, \widetilde{\sigma}_2(a_2)) = \widehat{u}_2(x, \widetilde{\sigma}_2(\cdot)),$$

where the strict inequality follows from the strict monotonicity of u_2^*. This contradicts the property whereby equilibrium strategies $\widehat{\sigma}_2(\cdot)$ and $\widetilde{\sigma}_2(\cdot)$ both maximize $\widehat{u}_2(x, \sigma_2(\cdot))$. Multiple equilibrium distributions over types being conceded to are in conflict with the requirement that types mimicked with a positive probability must have equal payoffs that are not smaller than the payoffs of the types that are not mimicked. Suppose, by contradiction, that there are two different equilibrium strategies for Player 2: $\sigma_2(a_2) \neq \sigma_2'(a_2)$ for some a_2. If $\sigma_2(a_2) > 0$ and

$\sigma_2'(a_2) > 0$, then $u_2(a_1, a_2, x, \sigma_2(a_2)) \neq u_2(a_1, a_2, x, \sigma_2'(a_2))$. Note that $u_2(a_1, a_2, x, \sigma_2(a_2)) = \widehat{u}_2(x, \sigma_2(\cdot))$ and $u_2(a_1, a_2, x, \sigma_2'(a_2)) = \widehat{u}_2(x, \sigma_2'(\cdot))$, , but $\widehat{u}_2(x, \sigma_2(\cdot)) \neq \widehat{u}_2(x, \sigma_2'(\cdot))$ contradicts the fact that $\sigma_2(\cdot)$ and $\sigma_2'(\cdot)$ both solve $\max_{\sigma_2(\cdot)\in\Delta_2(a_1,x)} \widehat{u}_2(x, \sigma_2(\cdot))$. If $\sigma_2(a_2)$ or $\sigma_2'(a_2)$ is zero, then by the first additional property of Player 2's equilibrium strategy above, there is an $a_2' > a_2$ such that $\sigma_2(a_2') > 0$, $\sigma_2'(a_2') > 0$, and $\sigma_2(a_2') \neq \sigma_2'(a_2')$, so that the contradiction arises again. Player 1 receives $u_1(a_1, x)$ in the equilibrium of the bargaining game $B(a_1, x)$. Thus, there exists a unique $\sigma_2(\cdot)$. □

Proof of Theorem 2. Denote by $u_1(a_1, x)$ the payoff of player 1 in the unique equilibrium of the bargaining game $B_1(a_1, x)$ with $A_1 = \{a_1\}$ and $|A_2| \geq 1$. The function $u_1(a_1, x)$ is continuous in x. Moreover, there exists $\underline{x}$ such that $u_1^*(a_1, x) = u_1^*(a_1, \underline{x})$ for all $x \leq \underline{x}$, and $u_1^*(a_1, x)$ is strictly increasing on $(\underline{x}, 1)$.

We characterize the equilibrium distribution σ_1 as the solution to

$$\max_{\sigma_1} \widetilde{u}(\sigma_1), \qquad \widetilde{u}(\sigma_1) = \min_{a_1 : \sigma_1(a_1)>0} u_1(a_1, x(\sigma_1(a_1))),$$

where

$$x(\sigma_1(a_1)) = \frac{z_1 \pi_1(a_1)}{z_1 \pi_1(a_1) + (1 - z_1)\sigma_1(a_1)}.$$

The continuity of $u_1(a_1, x)$ ensures the existence of an equilibrium, following the same fixed-point argument used above for the existence of σ_2 in the game $B(a_1, x)$.

Let $\overline{u}_1$ denote the maximized payoff. Clearly, $\overline{u}_1 \geq u_1(a_1, \underline{x})$ for all a_1. Let σ_1 and $\widehat{\sigma}_1$ be two equilibrium strategies of player 1. If $\overline{u}_1 > u_1(a_1, \underline{x})$, then by monotonicity of $u_1(a_1, x)$ we must have either $u_1(a_1, 1) > \overline{u}_1$, in which case the strict monotonicity of $u_1(a_1, x)$ on $(\underline{x}, 1)$ implies that there is a unique $x(\sigma_1(a_1))$ satisfying $u_1(a_1, x(\sigma_1(a_1))) = \overline{u}_1$ and hence $\sigma_1(a_1) = \widehat{\sigma}_1(a_1)$; or $u_1(a_1, 1) \leq \overline{u}_1$, in which case $u_1(a_1, x) < u_1(a_1, 1) \leq \overline{u}_1$ for all $x < 1$, implying $\sigma_1(a_1) = \widehat{\sigma}_1(a_1) = 0$.

Define

$$D_1 = \{a_1 \in A_1 : u_1(a_1, \underline{x}) = \overline{u}_1\}.$$

Since $\sigma_1(a_1) = \widehat{\sigma}_1(a_1)$ for all $a_1 \notin D_1$, it follows that

$$\sum_{a_1 \in D_1} \sigma_1(a_1) = \sum_{a_1 \in D_1} \widehat{\sigma}_1(a_1).$$

Let $A(\sigma_1)$ denote the probability that player 1 mimics some $a_1 \in D_1$ and agreement is reached

at time 0. Because $a_1 \in D_1$ implies $\sigma_2(\overline{a}_2 \mid a_1) = 1$, we must have $a_1 \geq 1 - \overline{a}_2$, and thus

$$\begin{aligned}
A(\sigma_1) &= \sum_{a_1 \in D_1} q_1(a_1, \overline{a}_2, x(\sigma_1(a_1)), 1)[1 - x(\sigma_1(a_1))]\,[z_1\pi_1(a_1) + (1 - z_1)\sigma_1(a_1)] \\
&= \sum_{a_1 \in D_1} \frac{K(a_1, \overline{a}_2, 1) - x(\sigma_1(a_1))}{K(a_1, \overline{a}_2, 1)}[z_1\pi_1(a_1) + (1 - z_1)\sigma_1(a_1)] \\
&= \sum_{a_1 \in D_1} (1 - z_1)\sigma_1(a_1) + \sum_{a_1 \in D_1} z_1\pi_1(a_1) - \sum_{a_1 \in D_1} \frac{z_1\pi_1(a_1)}{K(a_1, \overline{a}_2, 1)}.
\end{aligned}$$

Since the two equilibrium strategies assign the same total weight to D_1, we have $A(\sigma_1) = A(\widehat{\sigma}_1)$.

Finally, for any $a_1 \in D_1$, the probability that an unjustified player 1 mimics a_1 and does not concede at time 0 is

$$\begin{aligned}
\sigma_1(a_1)[1 - q_1(a_1, \overline{a}_2, x(\sigma_1(a_1)), 1)] &= \frac{\sigma_1(a_1)x(\sigma_1(a_1))}{1 - x(\sigma_1(a_1))} \frac{1 - K(a_1, \overline{a}_2, 1)}{K(a_1, \overline{a}_2, 1)} \\
&= \frac{\pi_1(a_1)z_1}{1 - z_1} \frac{1 - K(a_1, \overline{a}_2, 1)}{K(a_1, \overline{a}_2, 1)},
\end{aligned}$$

which is independent of σ_1. Therefore, both equilibrium strategies yield the same probabilities for mimicry and thus the same random outcome $\widetilde{\theta}$. □

Proof of Proposition 6

Proof. We derive the asymptotic expression for the exit time T_i using a singular perturbation argument with boundary-layer matching. Recall that $k_i(t) = 1 - \mu_i(t)$ satisfies

$$k_i'(t) = -\gamma_i k_i(t) - \frac{(1 - k_i(t))\big[r_j(1 - a_i) - \gamma_i k_i(t)\big(U_j(t) - (1 - a_i)\big)\big]}{a_i + a_j - 1},$$

with initial condition $k_i(0) = 1 - z_i \in (0, 1)$ and exit condition $k_i(T_i) = 0$.

We first consider the outer region, in which $k_i(t)$ is bounded away from zero. In this region, equilibrium feasibility requires $\lambda_i(t) > 0$. Since r_j is arbitrarily small while $k_i(t)$ is of order one, this condition implies that $U_j(t)$ must remain arbitrarily close to $1 - a_i$ throughout the outer region. Substituting $U_j(t) \approx 1 - a_i$ into the equation for k_i and neglecting terms of order r_j, the dynamics reduce to

$$k_i'(t) \approx -\gamma_i k_i(t).$$

By continuous dependence of solutions to ordinary differential equations on parameters, it fol-

lows that $k_i(t)$ is arbitrarily well approximated by

$$k_i(t) \approx k_i(0)e^{-\gamma_i t}$$

for all t in the outer region. Fix any ε satisfying $r_j \ll \varepsilon \ll 1$, and let t_1 denote the time at which $k_i(t_1) = \varepsilon$. The outer approximation then yields

$$t_1 = \frac{1}{\gamma_i} \ln\left(\frac{k_i(0)}{\varepsilon}\right),$$

where the approximation is arbitrarily accurate for r_j sufficiently small.

We next analyze the boundary layer in which $k_i(t)$ becomes of the same order as r_j. Introduce the inner time variable $s = t - t_1$ and rescale $k_i(t) = r_j\zeta(s)$, where $\zeta(s)$ is of order one. Upon entering the boundary layer, the matching requirement implies that $U_j(t)$ is arbitrarily close to $1-a_i$. Moreover, since the boundary-layer time scale is of order one and the drift of U_j is of order r_j, $U_j(t)$ remains arbitrarily close to $1 - a_i$ throughout the boundary layer. Substituting $k_i = r_j\zeta$ and $U_j = 1 - a_i$ into the exact equation and dividing through by r_j, we obtain the limiting inner equation

$$\zeta'(s) = -\gamma_i\zeta(s) - \frac{1 - a_i}{a_i + a_j - 1}.$$

The general solution is given by

$$\zeta(s) = \left(\zeta_0 + \frac{1 - a_i}{\gamma_i(a_i + a_j - 1)}\right) e^{-\gamma_i s} - \frac{1 - a_i}{\gamma_i(a_i + a_j - 1)},$$

where $\zeta_0 = \zeta(0)$.

Matching the inner and outer solutions requires that, as $s \to -\infty$, the inner expansion of $k_i(t) = r_j\zeta(s)$ coincides with the outer expansion evaluated at $t = t_1 + s$. Using the outer approximation $k_i(t_1 + s) \approx \varepsilon e^{-\gamma_i s}$, this condition implies

$$r_j\left(\zeta_0 + \frac{1 - a_i}{\gamma_i(a_i + a_j - 1)}\right) = \varepsilon.$$

The exit condition $k_i(T_i) = 0$ corresponds to $\zeta(s_f) = 0$, where $s_f = T_i - t_1$. Solving $\zeta(s_f) = 0$ yields

$$s_f = \frac{1}{\gamma_i} \ln\left(1 + \frac{\gamma_i(a_i + a_j - 1)}{1 - a_i}\,\zeta_0\right).$$

The total exit time is therefore $T_i = t_1 + s_f$. Substituting the expressions for t_1 and s_f, and

eliminating ζ_0 using the matching condition above, we obtain

$$T_i = \frac{1}{\gamma_i} \ln\left(\frac{k_i(0)\gamma_i(a_i + a_j - 1)}{(1 - a_i)r_j}\right).$$

The intermediate scale ε cancels out exactly, confirming the internal consistency of the matching procedure. Consequently, as r_j becomes arbitrarily small,

$$T_i = \frac{1}{\gamma_i} \ln\left(\frac{1}{r_j}\right) + \frac{1}{\gamma_i} \ln\left(\frac{(1 - z_i)\gamma_i(a_i + a_j - 1)}{1 - a_i}\right) + o(1),$$

where the approximation is arbitrarily good for r_j small and ε sufficiently larger than r_j. T_i grows logarithmically in $1/r_j$ with leading coefficient $1/\gamma_i$. Suppose that $r_1/r_2 = c \in (0, \infty)$. Substituting $r_1 = cr_2$ into the expression for T_2 yields

$$T_2 = \frac{1}{\gamma_2} \ln\left(\frac{1}{r_2}\right) - \frac{1}{\gamma_2} \ln c + \frac{1}{\gamma_2} \ln\left(\frac{k_2(0)\gamma_2(a_1 + a_2 - 1)}{1 - a_2}\right) + o(1).$$

Therefore, the difference between the two exit times satisfies

$$\begin{aligned} T_1 - T_2 = &\left(\frac{1}{\gamma_1} - \frac{1}{\gamma_2}\right) \ln\left(\frac{1}{r_2}\right) \\ &+ \left[\frac{1}{\gamma_1} \ln\left(\frac{k_1(0)\gamma_1(a_1 + a_2 - 1)}{1 - a_1}\right) - \frac{1}{\gamma_2} \ln\left(\frac{k_2(0)\gamma_2(a_1 + a_2 - 1)}{1 - a_2}\right) + \frac{1}{\gamma_2} \ln c\right] + o(1). \end{aligned}$$

Since $\ln(1/r_2)$ diverges, the sign of $T_1 - T_2$ is determined by the relative magnitudes of γ_1 and γ_2. In particular,

$$\gamma_1 > \gamma_2 \quad \Longleftrightarrow \quad T_1 < T_2.$$

Without loss of generality, assume that $\gamma_1 > \gamma_2$ and $r_1 = cr_2$, so Player 2 concedes with positive probability Q_2 at time zero. We now characterize $\widehat{Q}_2$ in (24). Recall that

$$\widehat{\lambda}_2 = \frac{r_1(1 - a_2)}{a_1 + a_2 - 1}.$$

Since $r_1 = cr_2$, it follows that $\widehat{\lambda}_2 = O(r_2)$. Therefore,

$$\widehat{\lambda}_2 T_1 = \frac{c(1 - a_2)}{a_1 + a_2 - 1} \cdot \frac{1}{\gamma_1} r_2 \ln\left(\frac{1}{r_2}\right) + O(r_2).$$

Because $r_2 \ln(1/r_2) \to 0$ as $r_2 \to 0$, we obtain $\widehat{\lambda}_2 T_1 \to 0$. Consequently,

$$\exp(\widehat{\lambda}_2 T_1) = 1 + \widehat{\lambda}_2 T_1 + o(\widehat{\lambda}_2 T_1).$$

Substituting this expansion into the expression

$$\widehat{Q}_2 = \frac{1 - z_2 \exp(\widehat{\lambda}_2 T_1)}{1 - z_2} = 1 - \frac{z_2}{1 - z_2} \widehat{\lambda}_2 T_1 + o(\widehat{\lambda}_2 T_1).$$

Since $\widehat{\lambda}_2 T_1 \to 0$, it follows that $\widehat{Q}_2$ converges to one. By construction, $\widehat{Q}_2 < Q_2 \leq 1$, so that Player 2 concedes almost immediately with probability arbitrarily close to one. By symmetry, when $\gamma_1 < \gamma_2$, the roles of the two players are reversed. Player 1 concedes at time zero with probability arbitrarily close to one.

When $\gamma_1 = \gamma_2 = \gamma$, the leading terms of the exit times are identical and of order $\gamma^{-1} \ln(1/r_j)$. Hence, a comparison of T_1 and T_2 requires keeping track of the constant terms in their asymptotic expansions. From the asymptotic characterization derived above, we can write

$$T_1 = \frac{1}{\gamma} \ln\left(\frac{1}{r_2}\right) + \beta_1 + o(1), \qquad T_2 = \frac{1}{\gamma} \ln\left(\frac{1}{r_2}\right) + \beta_2 + o(1),$$

where

$$\beta_1 \equiv \frac{1}{\gamma} \ln\left(\frac{(1 - z_1)\gamma(a_1 + a_2 - 1)}{1 - a_1}\right), \qquad \beta_2 \equiv \frac{1}{\gamma} \ln\left(\frac{(1 - z_2)\gamma(a_1 + a_2 - 1)}{1 - a_2}\right) - \frac{1}{\gamma} \ln c.$$

Subtracting the two expressions yields

$$T_1 - T_2 = \beta_1 - \beta_2 + o(1) = \frac{1}{\gamma} \ln\left(\frac{(1 - z_1)/(1 - a_1)}{(1 - z_2)/(1 - a_2)} \cdot c\right) + o(1).$$

It follows that, in the limit $r_1, r_2 \to 0$ with $r_1/r_2 \to c > 0$,

$$T_1 < T_2 \quad \Longleftrightarrow \quad \frac{(1 - z_1)/(1 - a_1)}{(1 - z_2)/(1 - a_2)} \cdot c < 1 \quad \Longleftrightarrow \quad \frac{\widehat{\lambda}_2}{(1 - z_2)} < \frac{\widehat{\lambda}_1}{(1 - z_1)}.$$

The identity of the player who concedes at time zero, as well as the associated concession mass, is determined by the interaction between initial reputations, demands, and patience. By the preceding arguments, whenever $T_1 < T_2$, equivalently $\frac{\widehat{\lambda}_2}{1-z_2} < \frac{\widehat{\lambda}_1}{1-z_1}$, Player 2 concedes at time zero with probability arbitrarily close to one. Conversely, Player 1 concedes at time zero with probability arbitrarily close to one. □